Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 13:09:29

Measurement Details:

- **Sample:** Muestra1-253c_agit
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

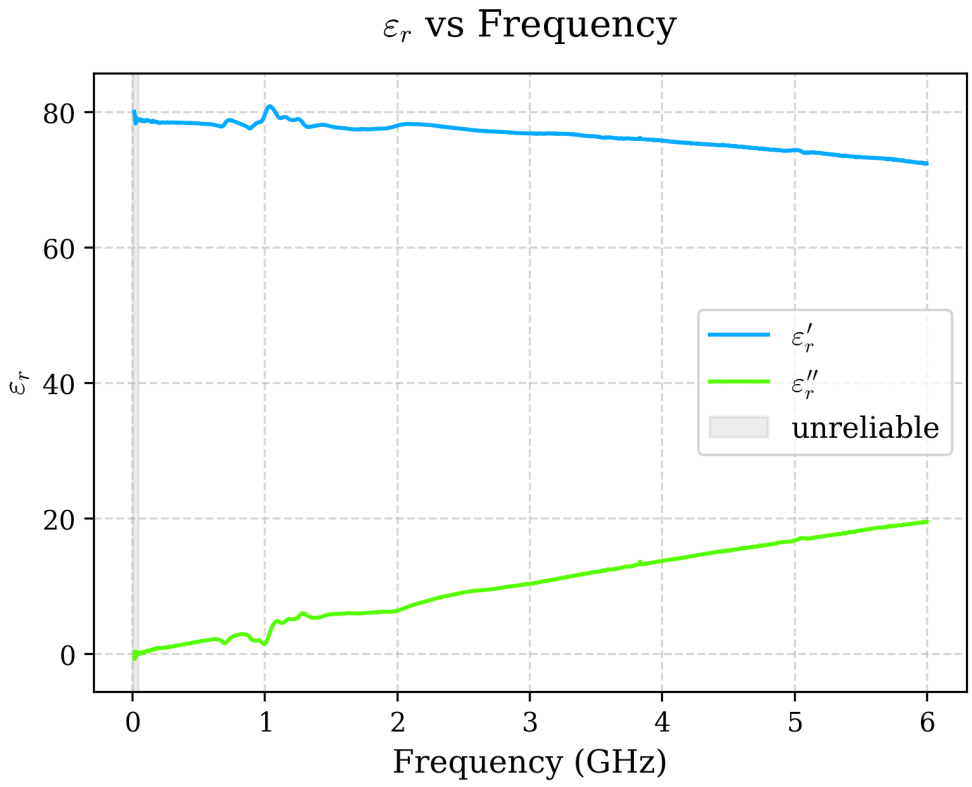
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table). Non-physical points ($\varepsilon_r'' < 0$ or $\varepsilon_r' < 1$): 5 of 1200, kept and marked † in the table, not discarded.

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	80.0333	-0.7024†	-0.0088
0.020	79.4609	0.3644	0.0046
0.025	78.3076	-0.5450†	-0.0070
0.030	78.7720	-0.0832†	-0.0011
0.035	79.0796	0.0120	0.0002
0.040	78.8404	0.2356	0.0030
0.045	78.7922	-0.0181†	-0.0002
0.050	78.8452	0.0852	0.0011
0.055	78.8732	-0.0324†	-0.0004
0.060	78.7103	0.1712	0.0022
0.065	78.9224	0.2178	0.0028
0.070	78.7680	0.0818	0.0010
0.075	78.6354	0.2723	0.0035
0.080	78.7502	0.1832	0.0023

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.085	78.6807	0.2264	0.0029
0.090	78.7762	0.3324	0.0042
0.095	78.6228	0.2036	0.0026
0.100	78.7459	0.3494	0.0044
0.105	78.7895	0.3932	0.0050
0.110	78.8410	0.4498	0.0057
0.115	78.7089	0.5066	0.0064
0.120	78.7637	0.3705	0.0047
0.125	78.7456	0.4041	0.0051
0.130	78.6807	0.4588	0.0058
0.135	78.6171	0.5396	0.0069
0.140	78.6323	0.5745	0.0073
0.145	78.5292	0.6218	0.0079
0.150	78.7033	0.6356	0.0081
0.155	78.5356	0.6103	0.0078
0.160	78.6774	0.7217	0.0092
0.165	78.5877	0.7388	0.0094
0.170	78.5523	0.6458	0.0082
0.175	78.5788	0.8315	0.0106
0.180	78.5735	0.8049	0.0102
0.185	78.5304	0.8011	0.0102
0.190	78.4600	0.8698	0.0111
0.195	78.4917	0.8810	0.0112
0.200	78.4147	0.8730	0.0111
0.205	78.3969	0.8075	0.0103
0.210	78.3833	0.8920	0.0114
0.215	78.4618	0.8718	0.0111
0.220	78.4686	0.8761	0.0112
0.225	78.4697	0.8542	0.0109
0.230	78.4619	0.8849	0.0113
0.235	78.4624	0.9148	0.0117
0.240	78.4448	0.9364	0.0119
0.245	78.4451	0.9220	0.0118
0.250	78.4342	0.9624	0.0123
0.255	78.4377	0.9121	0.0116
0.260	78.4807	0.9481	0.0121
0.265	78.4761	0.9441	0.0120
0.270	78.4542	1.0141	0.0129
0.275	78.4162	0.9841	0.0125
0.280	78.4681	1.0661	0.0136
0.285	78.4376	1.0327	0.0132
0.290	78.4467	1.0642	0.0136
0.295	78.4323	1.0690	0.0136
0.300	78.4807	1.0648	0.0136
0.305	78.4503	1.1285	0.0144
0.310	78.4446	1.1449	0.0146

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.315	78.4471	1.1577	0.0148
0.320	78.4077	1.2085	0.0154
0.325	78.4500	1.1555	0.0147
0.330	78.4612	1.2334	0.0157
0.335	78.4353	1.2296	0.0157
0.340	78.3781	1.2359	0.0158
0.345	78.4509	1.2433	0.0158
0.350	78.4294	1.2910	0.0165
0.355	78.4339	1.2833	0.0164
0.360	78.4378	1.3010	0.0166
0.365	78.4157	1.3121	0.0167
0.370	78.4435	1.3296	0.0169
0.375	78.4181	1.3601	0.0173
0.380	78.4026	1.3953	0.0178
0.385	78.4190	1.4027	0.0179
0.390	78.4104	1.3667	0.0174
0.395	78.3888	1.4044	0.0179
0.400	78.3952	1.4696	0.0187
0.405	78.3849	1.4751	0.0188
0.410	78.3898	1.4908	0.0190
0.415	78.3758	1.4881	0.0190
0.420	78.3645	1.5207	0.0194
0.425	78.4059	1.5397	0.0196
0.430	78.3982	1.5236	0.0194
0.435	78.3484	1.5530	0.0198
0.440	78.3728	1.5534	0.0198
0.445	78.4089	1.5780	0.0201
0.450	78.3744	1.6398	0.0209
0.455	78.3599	1.6420	0.0210
0.460	78.3621	1.6672	0.0213
0.465	78.3536	1.6866	0.0215
0.470	78.3740	1.6921	0.0216
0.475	78.3568	1.7054	0.0218
0.480	78.3298	1.7185	0.0219
0.485	78.3608	1.7475	0.0223
0.490	78.3188	1.7717	0.0226
0.495	78.3600	1.7903	0.0228
0.500	78.3192	1.7934	0.0229
0.505	78.3108	1.8134	0.0232
0.510	78.3100	1.8313	0.0234
0.515	78.3112	1.8443	0.0236
0.520	78.2994	1.8642	0.0238
0.525	78.2490	1.8882	0.0241
0.530	78.2876	1.9114	0.0244
0.535	78.2631	1.9096	0.0244
0.540	78.2758	1.9353	0.0247

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.545	78.2586	1.9528	0.0250
0.550	78.2295	1.9623	0.0251
0.555	78.2504	1.9764	0.0253
0.560	78.2378	1.9891	0.0254
0.565	78.2349	2.0111	0.0257
0.570	78.2143	2.0124	0.0257
0.575	78.2326	2.0362	0.0260
0.580	78.2010	2.0549	0.0263
0.585	78.2113	2.0520	0.0262
0.590	78.2093	2.0686	0.0264
0.595	78.1992	2.1114	0.0270
0.600	78.1919	2.1270	0.0272
0.605	78.1762	2.1464	0.0275
0.610	78.1421	2.1624	0.0277
0.615	78.1190	2.1645	0.0277
0.620	78.1144	2.1852	0.0280
0.625	78.0772	2.1836	0.0280
0.630	78.0365	2.1675	0.0278
0.635	78.0204	2.1772	0.0279
0.640	78.0103	2.1506	0.0276
0.645	77.9788	2.1632	0.0277
0.650	77.9760	2.1381	0.0274
0.655	77.9311	2.0830	0.0267
0.660	77.9077	2.0679	0.0265
0.665	77.8893	2.0145	0.0259
0.670	77.8597	1.9682	0.0253
0.675	77.8742	1.8798	0.0241
0.680	77.9049	1.8114	0.0233
0.685	77.9338	1.7109	0.0220
0.690	78.0369	1.6461	0.0211
0.695	78.1556	1.5748	0.0202
0.700	78.3111	1.5850	0.0202
0.705	78.4576	1.6117	0.0205
0.710	78.6016	1.6732	0.0213
0.715	78.6923	1.7684	0.0225
0.720	78.7589	1.8909	0.0240
0.725	78.7961	2.0143	0.0256
0.730	78.8105	2.1203	0.0269
0.735	78.8313	2.2009	0.0279
0.740	78.7822	2.3148	0.0294
0.745	78.7753	2.3959	0.0304
0.750	78.7433	2.4450	0.0311
0.755	78.7389	2.5401	0.0323
0.760	78.6890	2.5741	0.0327
0.765	78.6472	2.6325	0.0335
0.770	78.6075	2.6630	0.0339

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.775	78.5633	2.6851	0.0342
0.780	78.5470	2.7317	0.0348
0.785	78.5016	2.7637	0.0352
0.790	78.4592	2.7850	0.0355
0.795	78.4190	2.8092	0.0358
0.800	78.4067	2.8370	0.0362
0.805	78.3731	2.8411	0.0363
0.810	78.3300	2.8682	0.0366
0.815	78.2953	2.8809	0.0368
0.820	78.2305	2.8995	0.0371
0.825	78.1894	2.9143	0.0373
0.830	78.1256	2.9058	0.0372
0.835	78.1038	2.9126	0.0373
0.840	78.0642	2.8751	0.0368
0.845	78.0358	2.8882	0.0370
0.850	77.9828	2.8730	0.0368
0.855	77.9375	2.8582	0.0367
0.860	77.8848	2.8461	0.0365
0.865	77.8370	2.8239	0.0363
0.870	77.7609	2.7944	0.0359
0.875	77.6982	2.7164	0.0350
0.880	77.6277	2.6465	0.0341
0.885	77.6003	2.5126	0.0324
0.890	77.5842	2.3986	0.0309
0.895	77.6357	2.2474	0.0289
0.900	77.6934	2.1376	0.0275
0.905	77.8162	2.0582	0.0264
0.910	77.8921	2.0295	0.0261
0.915	77.9867	1.9827	0.0254
0.920	78.0422	1.9758	0.0253
0.925	78.1356	1.9356	0.0248
0.930	78.2396	1.9280	0.0246
0.935	78.3251	1.9303	0.0246
0.940	78.4005	1.9608	0.0250
0.945	78.4306	1.9892	0.0254
0.950	78.4580	2.0138	0.0257
0.955	78.4630	2.0008	0.0255
0.960	78.4813	1.9646	0.0250
0.965	78.5324	1.9158	0.0244
0.970	78.5620	1.8672	0.0238
0.975	78.6212	1.7605	0.0224
0.980	78.7219	1.6714	0.0212
0.985	78.8742	1.5757	0.0200
0.990	79.0787	1.5150	0.0192
0.995	79.2809	1.4558	0.0184
1.000	79.5587	1.4808	0.0186

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.005	79.8440	1.5488	0.0194
1.010	80.0857	1.6856	0.0210
1.015	80.3227	1.8680	0.0233
1.020	80.5365	2.0983	0.0261
1.025	80.6833	2.3564	0.0292
1.030	80.8089	2.6424	0.0327
1.035	80.8589	2.9293	0.0362
1.040	80.8640	3.2181	0.0398
1.045	80.8311	3.4884	0.0432
1.050	80.7562	3.7440	0.0464
1.055	80.6596	3.9878	0.0494
1.060	80.5436	4.1403	0.0514
1.065	80.4090	4.3183	0.0537
1.070	80.2994	4.4529	0.0555
1.075	80.1667	4.5921	0.0573
1.080	80.0093	4.7223	0.0590
1.085	79.8276	4.8113	0.0603
1.090	79.6475	4.8476	0.0609
1.095	79.4837	4.8743	0.0613
1.100	79.3398	4.8301	0.0609
1.105	79.2288	4.7706	0.0602
1.110	79.1497	4.6951	0.0593
1.115	79.1144	4.6559	0.0589
1.120	79.1068	4.6019	0.0582
1.125	79.1327	4.5733	0.0578
1.130	79.1441	4.5529	0.0575
1.135	79.2238	4.5550	0.0575
1.140	79.2496	4.6167	0.0583
1.145	79.2544	4.6691	0.0589
1.150	79.2478	4.7250	0.0596
1.155	79.2722	4.8144	0.0607
1.160	79.2592	4.8959	0.0618
1.165	79.2093	4.9774	0.0628
1.170	79.1554	5.0402	0.0637
1.175	79.0916	5.1061	0.0646
1.180	79.0180	5.1314	0.0649
1.185	78.9296	5.1606	0.0654
1.190	78.8830	5.1389	0.0651
1.195	78.8384	5.1118	0.0648
1.200	78.8298	5.1011	0.0647
1.205	78.8252	5.1053	0.0648
1.210	78.8046	5.0997	0.0647
1.215	78.8393	5.1054	0.0648
1.220	78.8400	5.1109	0.0648
1.225	78.8271	5.1245	0.0650
1.230	78.8578	5.1498	0.0653

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.235	78.8773	5.1779	0.0656
1.240	78.9209	5.2063	0.0660
1.245	78.9567	5.2699	0.0667
1.250	78.9671	5.3663	0.0680
1.255	78.9485	5.4857	0.0695
1.260	78.9266	5.5824	0.0707
1.265	78.9011	5.6801	0.0720
1.270	78.7952	5.7938	0.0735
1.275	78.7088	5.8792	0.0747
1.280	78.6006	5.9569	0.0758
1.285	78.4605	5.9741	0.0761
1.290	78.3156	6.0076	0.0767
1.295	78.1875	5.9614	0.0762
1.300	78.0969	5.9283	0.0759
1.305	77.9958	5.8648	0.0752
1.310	77.9220	5.7975	0.0744
1.315	77.8709	5.7237	0.0735
1.320	77.8245	5.6548	0.0727
1.325	77.7994	5.6016	0.0720
1.330	77.8166	5.5407	0.0712
1.335	77.8005	5.4883	0.0705
1.340	77.7953	5.4549	0.0701
1.345	77.8274	5.3949	0.0693
1.350	77.8448	5.3998	0.0694
1.355	77.8455	5.3684	0.0690
1.360	77.8889	5.3434	0.0686
1.365	77.8985	5.3210	0.0683
1.370	77.9201	5.3252	0.0683
1.375	77.9384	5.3362	0.0685
1.380	77.9676	5.3333	0.0684
1.385	77.9809	5.3444	0.0685
1.390	77.9913	5.3468	0.0686
1.395	78.0087	5.3328	0.0684
1.400	78.0212	5.3549	0.0686
1.405	78.0402	5.3705	0.0688
1.410	78.0476	5.3839	0.0690
1.415	78.0728	5.4074	0.0693
1.420	78.0935	5.4170	0.0694
1.425	78.0954	5.4467	0.0697
1.430	78.1102	5.4755	0.0701
1.435	78.0987	5.5298	0.0708
1.440	78.1143	5.5485	0.0710
1.445	78.1306	5.5977	0.0716
1.450	78.1122	5.6188	0.0719
1.455	78.0762	5.6630	0.0725
1.460	78.0750	5.6923	0.0729

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.465	78.0624	5.7067	0.0731
1.470	78.0403	5.7518	0.0737
1.475	78.0149	5.7642	0.0739
1.480	77.9763	5.7883	0.0742
1.485	77.9577	5.7983	0.0744
1.490	77.9361	5.8268	0.0748
1.495	77.8961	5.8348	0.0749
1.500	77.8799	5.8534	0.0752
1.505	77.8735	5.8698	0.0754
1.510	77.8404	5.8693	0.0754
1.515	77.8254	5.8763	0.0755
1.520	77.8150	5.8831	0.0756
1.525	77.7813	5.8991	0.0758
1.530	77.7579	5.8919	0.0758
1.535	77.7355	5.9168	0.0761
1.540	77.7295	5.8987	0.0759
1.545	77.7326	5.9009	0.0759
1.550	77.7143	5.9000	0.0759
1.555	77.6807	5.8866	0.0758
1.560	77.6827	5.9179	0.0762
1.565	77.6713	5.8973	0.0759
1.570	77.6613	5.9047	0.0760
1.575	77.6576	5.8936	0.0759
1.580	77.6683	5.9289	0.0763
1.585	77.6299	5.9455	0.0766
1.590	77.6443	5.9434	0.0765
1.595	77.6249	5.9350	0.0765
1.600	77.6261	5.9503	0.0767
1.605	77.6125	5.9424	0.0766
1.610	77.5956	5.9599	0.0768
1.615	77.5882	5.9761	0.0770
1.620	77.5810	6.0019	0.0774
1.625	77.5571	5.9938	0.0773
1.630	77.5422	6.0057	0.0775
1.635	77.5273	6.0203	0.0777
1.640	77.5105	5.9915	0.0773
1.645	77.4986	6.0051	0.0775
1.650	77.4852	6.0096	0.0776
1.655	77.4644	6.0033	0.0775
1.660	77.4582	5.9849	0.0773
1.665	77.4605	5.9721	0.0771
1.670	77.4456	6.0107	0.0776
1.675	77.4391	5.9941	0.0774
1.680	77.4354	5.9810	0.0772
1.685	77.4333	5.9740	0.0771
1.690	77.4364	5.9683	0.0771

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.695	77.4402	5.9675	0.0771
1.700	77.4176	5.9722	0.0771
1.705	77.4345	5.9630	0.0770
1.710	77.4442	5.9621	0.0770
1.715	77.4331	5.9684	0.0771
1.720	77.4698	5.9765	0.0771
1.725	77.4373	5.9749	0.0772
1.730	77.4828	6.0045	0.0775
1.735	77.4662	5.9727	0.0771
1.740	77.4613	5.9942	0.0774
1.745	77.4808	6.0064	0.0775
1.750	77.4772	6.0208	0.0777
1.755	77.4752	6.0332	0.0779
1.760	77.4689	6.0280	0.0778
1.765	77.4731	6.0266	0.0778
1.770	77.4510	6.0577	0.0782
1.775	77.4522	6.0653	0.0783
1.780	77.4712	6.0772	0.0784
1.785	77.4453	6.0774	0.0785
1.790	77.4435	6.0741	0.0784
1.795	77.4560	6.0901	0.0786
1.800	77.4632	6.1083	0.0789
1.805	77.4810	6.0898	0.0786
1.810	77.4653	6.1000	0.0787
1.815	77.4699	6.1077	0.0788
1.820	77.4808	6.1087	0.0788
1.825	77.4950	6.1052	0.0788
1.830	77.4974	6.1175	0.0789
1.835	77.4960	6.1290	0.0791
1.840	77.5160	6.1332	0.0791
1.845	77.5195	6.1918	0.0799
1.850	77.5318	6.1327	0.0791
1.855	77.5309	6.1701	0.0796
1.860	77.5490	6.1458	0.0793
1.865	77.5384	6.1979	0.0799
1.870	77.5516	6.1911	0.0798
1.875	77.5395	6.1981	0.0799
1.880	77.5435	6.1956	0.0799
1.885	77.5340	6.2225	0.0803
1.890	77.5592	6.2086	0.0800
1.895	77.5593	6.2116	0.0801
1.900	77.5632	6.2099	0.0801
1.905	77.5699	6.2125	0.0801
1.910	77.5727	6.2135	0.0801
1.915	77.5841	6.2249	0.0802
1.920	77.6152	6.2140	0.0801

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.925	77.6083	6.1956	0.0798
1.930	77.6344	6.2277	0.0802
1.935	77.6551	6.2060	0.0799
1.940	77.6906	6.2197	0.0801
1.945	77.7082	6.2130	0.0800
1.950	77.7186	6.2109	0.0799
1.955	77.7337	6.2214	0.0800
1.960	77.7606	6.2237	0.0800
1.965	77.7891	6.2482	0.0803
1.970	77.8092	6.2409	0.0802
1.975	77.8519	6.2671	0.0805
1.980	77.8744	6.2862	0.0807
1.985	77.9092	6.2888	0.0807
1.990	77.9294	6.3348	0.0813
1.995	77.9519	6.3425	0.0814
2.000	77.9904	6.3501	0.0814
2.005	78.0159	6.3882	0.0819
2.010	78.0272	6.4012	0.0820
2.015	78.0640	6.4545	0.0827
2.020	78.1109	6.4813	0.0830
2.025	78.1098	6.5289	0.0836
2.030	78.1123	6.5620	0.0840
2.035	78.1638	6.5932	0.0844
2.040	78.1663	6.6259	0.0848
2.045	78.1822	6.6648	0.0852
2.050	78.2009	6.7001	0.0857
2.055	78.2007	6.7552	0.0864
2.060	78.2174	6.7810	0.0867
2.065	78.2309	6.8290	0.0873
2.070	78.2428	6.8655	0.0877
2.075	78.2333	6.8944	0.0881
2.080	78.2227	6.9474	0.0888
2.085	78.2315	6.9879	0.0893
2.090	78.2339	7.0227	0.0898
2.095	78.2428	7.0527	0.0901
2.100	78.2107	7.0963	0.0907
2.105	78.2211	7.1148	0.0910
2.110	78.2145	7.1751	0.0917
2.115	78.2160	7.1682	0.0916
2.120	78.2322	7.1993	0.0920
2.125	78.2218	7.2305	0.0924
2.130	78.1934	7.2561	0.0928
2.135	78.2092	7.2880	0.0932
2.140	78.2187	7.3344	0.0938
2.145	78.2057	7.3710	0.0943
2.150	78.2084	7.3976	0.0946

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.155	78.2207	7.4179	0.0948
2.160	78.1994	7.4561	0.0953
2.165	78.1897	7.4803	0.0957
2.170	78.1887	7.4993	0.0959
2.175	78.1814	7.5327	0.0963
2.180	78.1731	7.5546	0.0966
2.185	78.1659	7.5849	0.0970
2.190	78.1650	7.6177	0.0975
2.195	78.1521	7.6492	0.0979
2.200	78.1593	7.6668	0.0981
2.205	78.1403	7.6957	0.0985
2.210	78.1438	7.7321	0.0989
2.215	78.1189	7.7573	0.0993
2.220	78.1228	7.7843	0.0996
2.225	78.1127	7.7914	0.0997
2.230	78.0784	7.8273	0.1002
2.235	78.0920	7.8457	0.1005
2.240	78.0728	7.9005	0.1012
2.245	78.0759	7.9145	0.1014
2.250	78.0861	7.9571	0.1019
2.255	78.0672	7.9543	0.1019
2.260	78.0546	7.9890	0.1024
2.265	78.0346	8.0226	0.1028
2.270	78.0412	8.0605	0.1033
2.275	78.0346	8.0657	0.1034
2.280	78.0105	8.1004	0.1038
2.285	77.9857	8.1470	0.1045
2.290	77.9953	8.1585	0.1046
2.295	77.9727	8.1929	0.1051
2.300	77.9702	8.2288	0.1055
2.305	77.9588	8.2525	0.1059
2.310	77.9179	8.2917	0.1064
2.315	77.9304	8.3002	0.1065
2.320	77.8925	8.3244	0.1069
2.325	77.8932	8.3375	0.1070
2.330	77.8934	8.3492	0.1072
2.335	77.8467	8.3875	0.1077
2.340	77.8669	8.4003	0.1079
2.345	77.8339	8.4142	0.1081
2.350	77.8291	8.4498	0.1086
2.355	77.8175	8.4744	0.1089
2.360	77.7935	8.4798	0.1090
2.365	77.7937	8.5143	0.1094
2.370	77.7836	8.5114	0.1094
2.375	77.7622	8.5504	0.1100
2.380	77.7672	8.5679	0.1102

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.385	77.7437	8.5926	0.1105
2.390	77.7256	8.6116	0.1108
2.395	77.7403	8.6265	0.1110
2.400	77.7340	8.6561	0.1114
2.405	77.7034	8.6894	0.1118
2.410	77.6982	8.6987	0.1120
2.415	77.6791	8.7228	0.1123
2.420	77.6690	8.7205	0.1123
2.425	77.6677	8.7777	0.1130
2.430	77.6457	8.7636	0.1129
2.435	77.6372	8.8014	0.1134
2.440	77.6328	8.8422	0.1139
2.445	77.6219	8.8417	0.1139
2.450	77.6116	8.8702	0.1143
2.455	77.5914	8.8783	0.1144
2.460	77.5961	8.9105	0.1148
2.465	77.5790	8.9173	0.1149
2.470	77.5679	8.9613	0.1155
2.475	77.5711	8.9668	0.1156
2.480	77.5370	8.9965	0.1160
2.485	77.5386	9.0178	0.1163
2.490	77.5259	9.0141	0.1163
2.495	77.5358	9.0599	0.1168
2.500	77.5009	9.0768	0.1171
2.505	77.4960	9.1078	0.1175
2.510	77.4881	9.1137	0.1176
2.515	77.4841	9.1134	0.1176
2.520	77.4667	9.1354	0.1179
2.525	77.4492	9.1580	0.1182
2.530	77.4385	9.1752	0.1185
2.535	77.4220	9.1840	0.1186
2.540	77.4108	9.1946	0.1188
2.545	77.4232	9.2101	0.1190
2.550	77.4009	9.2160	0.1191
2.555	77.3819	9.2484	0.1195
2.560	77.3725	9.2566	0.1196
2.565	77.3656	9.2768	0.1199
2.570	77.3535	9.2617	0.1197
2.575	77.3335	9.2924	0.1202
2.580	77.3288	9.3069	0.1204
2.585	77.3095	9.3054	0.1204
2.590	77.3024	9.3272	0.1207
2.595	77.3086	9.3175	0.1205
2.600	77.2894	9.3353	0.1208
2.605	77.2633	9.3313	0.1208
2.610	77.2565	9.3499	0.1210

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.615	77.2388	9.3800	0.1214
2.620	77.2490	9.3785	0.1214
2.625	77.2260	9.3729	0.1214
2.630	77.2329	9.3844	0.1215
2.635	77.2197	9.3934	0.1216
2.640	77.2229	9.4047	0.1218
2.645	77.2290	9.4034	0.1218
2.650	77.2166	9.4161	0.1219
2.655	77.1974	9.4473	0.1224
2.660	77.2052	9.4323	0.1222
2.665	77.2152	9.4513	0.1224
2.670	77.1635	9.4473	0.1224
2.675	77.2002	9.4605	0.1225
2.680	77.1799	9.4775	0.1228
2.685	77.1746	9.4817	0.1229
2.690	77.1660	9.4881	0.1230
2.695	77.1564	9.5066	0.1232
2.700	77.1568	9.5082	0.1232
2.705	77.1389	9.5377	0.1236
2.710	77.1620	9.5374	0.1236
2.715	77.1501	9.5419	0.1237
2.720	77.1600	9.5713	0.1240
2.725	77.1139	9.5863	0.1243
2.730	77.1332	9.5803	0.1242
2.735	77.1132	9.6058	0.1246
2.740	77.1192	9.6334	0.1249
2.745	77.0923	9.6305	0.1249
2.750	77.1220	9.6584	0.1252
2.755	77.1145	9.6432	0.1250
2.760	77.0897	9.6687	0.1254
2.765	77.0772	9.6962	0.1258
2.770	77.0558	9.7095	0.1260
2.775	77.0814	9.7123	0.1260
2.780	77.0647	9.7260	0.1262
2.785	77.0761	9.7451	0.1264
2.790	77.0464	9.7567	0.1266
2.795	77.0523	9.7961	0.1271
2.800	77.0699	9.8041	0.1272
2.805	77.0327	9.8226	0.1275
2.810	77.0438	9.8247	0.1275
2.815	77.0349	9.8668	0.1281
2.820	77.0302	9.8719	0.1282
2.825	77.0235	9.8625	0.1280
2.830	76.9921	9.9102	0.1287
2.835	77.0148	9.9042	0.1286
2.840	77.0178	9.9301	0.1289

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.845	76.9996	9.9120	0.1287
2.850	76.9927	9.9384	0.1291
2.855	76.9651	9.9601	0.1294
2.860	76.9785	9.9703	0.1295
2.865	76.9637	9.9696	0.1295
2.870	76.9569	9.9792	0.1297
2.875	76.9713	10.0060	0.1300
2.880	76.9628	10.0087	0.1300
2.885	76.9162	10.0359	0.1305
2.890	76.9198	10.0340	0.1304
2.895	76.9050	10.0799	0.1311
2.900	76.9089	10.0769	0.1310
2.905	76.8954	10.0868	0.1312
2.910	76.9204	10.1062	0.1314
2.915	76.9102	10.1215	0.1316
2.920	76.8946	10.1383	0.1318
2.925	76.8927	10.1551	0.1321
2.930	76.8769	10.1948	0.1326
2.935	76.8851	10.1804	0.1324
2.940	76.8737	10.2016	0.1327
2.945	76.8723	10.2128	0.1329
2.950	76.8799	10.1855	0.1325
2.955	76.8663	10.2173	0.1329
2.960	76.8860	10.2430	0.1332
2.965	76.8571	10.2537	0.1334
2.970	76.8624	10.2684	0.1336
2.975	76.8599	10.2597	0.1335
2.980	76.8634	10.2763	0.1337
2.985	76.8445	10.3154	0.1342
2.990	76.8613	10.3081	0.1341
2.995	76.8401	10.2969	0.1340
3.000	76.8345	10.3499	0.1347
3.005	76.8283	10.3740	0.1350
3.010	76.8465	10.3835	0.1351
3.015	76.8451	10.3791	0.1351
3.020	76.8249	10.4096	0.1355
3.025	76.8783	10.4085	0.1354
3.030	76.8190	10.4141	0.1356
3.035	76.8136	10.4245	0.1357
3.040	76.8358	10.4688	0.1362
3.045	76.8373	10.4517	0.1360
3.050	76.7671	10.5127	0.1369
3.055	76.7991	10.4848	0.1365
3.060	76.8145	10.5381	0.1372
3.065	76.7877	10.5503	0.1374
3.070	76.8494	10.5698	0.1375

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.075	76.8159	10.6184	0.1382
3.080	76.8329	10.6236	0.1383
3.085	76.8558	10.6299	0.1383
3.090	76.8124	10.6475	0.1386
3.095	76.7975	10.6801	0.1391
3.100	76.8242	10.6944	0.1392
3.105	76.7933	10.7341	0.1398
3.110	76.7978	10.7543	0.1400
3.115	76.8305	10.7466	0.1399
3.120	76.8080	10.7486	0.1399
3.125	76.8383	10.7741	0.1402
3.130	76.8287	10.7473	0.1399
3.135	76.8533	10.7911	0.1404
3.140	76.8420	10.8309	0.1410
3.145	76.8196	10.8048	0.1407
3.150	76.8605	10.8499	0.1412
3.155	76.8332	10.8789	0.1416
3.160	76.8548	10.9009	0.1418
3.165	76.8329	10.8863	0.1417
3.170	76.8174	10.8910	0.1418
3.175	76.8182	10.9494	0.1425
3.180	76.8627	10.9654	0.1427
3.185	76.8166	10.9621	0.1427
3.190	76.8073	10.9947	0.1431
3.195	76.8509	11.0118	0.1433
3.200	76.7693	11.0564	0.1440
3.205	76.8030	11.0286	0.1436
3.210	76.8029	11.0919	0.1444
3.215	76.8086	11.0985	0.1445
3.220	76.7595	11.1414	0.1451
3.225	76.8087	11.1280	0.1449
3.230	76.7998	11.1470	0.1451
3.235	76.7961	11.1804	0.1456
3.240	76.7921	11.1941	0.1458
3.245	76.8043	11.2268	0.1462
3.250	76.7740	11.2149	0.1461
3.255	76.8076	11.2697	0.1467
3.260	76.8014	11.2675	0.1467
3.265	76.7699	11.2856	0.1470
3.270	76.7537	11.3122	0.1474
3.275	76.7760	11.3215	0.1475
3.280	76.7749	11.3225	0.1475
3.285	76.8070	11.3966	0.1484
3.290	76.7703	11.3410	0.1477
3.295	76.7357	11.3816	0.1483
3.300	76.7544	11.4191	0.1488

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.305	76.7620	11.4311	0.1489
3.310	76.7251	11.4419	0.1491
3.315	76.7046	11.4471	0.1492
3.320	76.7257	11.4689	0.1495
3.325	76.6761	11.4449	0.1493
3.330	76.7064	11.5628	0.1507
3.335	76.7448	11.5610	0.1506
3.340	76.6867	11.5484	0.1506
3.345	76.6674	11.5974	0.1513
3.350	76.6423	11.5911	0.1512
3.355	76.6630	11.6137	0.1515
3.360	76.6431	11.6497	0.1520
3.365	76.6733	11.6384	0.1518
3.370	76.6206	11.6655	0.1523
3.375	76.6159	11.6912	0.1526
3.380	76.5692	11.7013	0.1528
3.385	76.5637	11.6783	0.1525
3.390	76.5711	11.7396	0.1533
3.395	76.5777	11.7205	0.1531
3.400	76.5348	11.7382	0.1534
3.405	76.5133	11.7997	0.1542
3.410	76.5111	11.7895	0.1541
3.415	76.5149	11.8182	0.1545
3.420	76.4932	11.8497	0.1549
3.425	76.4963	11.8239	0.1546
3.430	76.4968	11.8633	0.1551
3.435	76.4855	11.8953	0.1555
3.440	76.4734	11.9025	0.1556
3.445	76.4311	11.8728	0.1553
3.450	76.4236	11.9199	0.1560
3.455	76.4617	11.9581	0.1564
3.460	76.4747	11.9787	0.1566
3.465	76.4198	12.0432	0.1576
3.470	76.4170	12.0014	0.1571
3.475	76.4328	12.0231	0.1573
3.480	76.4350	12.0383	0.1575
3.485	76.4160	12.0921	0.1582
3.490	76.4163	12.0700	0.1580
3.495	76.3971	12.0891	0.1582
3.500	76.4085	12.0893	0.1582
3.505	76.4059	12.0762	0.1581
3.510	76.3536	12.1152	0.1587
3.515	76.3324	12.1131	0.1587
3.520	76.3476	12.1735	0.1594
3.525	76.3767	12.1533	0.1591
3.530	76.3133	12.2075	0.1600

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.535	76.2946	12.1775	0.1596
3.540	76.2532	12.2221	0.1603
3.545	76.2261	12.1941	0.1600
3.550	76.2006	12.2934	0.1613
3.555	76.2300	12.2586	0.1608
3.560	76.2185	12.3222	0.1617
3.565	76.2053	12.3675	0.1623
3.570	76.2431	12.3420	0.1619
3.575	76.2833	12.2763	0.1609
3.580	76.2692	12.3835	0.1624
3.585	76.2007	12.4478	0.1634
3.590	76.2461	12.4690	0.1635
3.595	76.2685	12.4633	0.1634
3.600	76.2649	12.4727	0.1635
3.605	76.2226	12.4607	0.1635
3.610	76.2474	12.4809	0.1637
3.615	76.2477	12.4540	0.1633
3.620	76.2362	12.4302	0.1630
3.625	76.1955	12.4999	0.1641
3.630	76.1984	12.4678	0.1636
3.635	76.1891	12.4751	0.1637
3.640	76.1560	12.4723	0.1638
3.645	76.1678	12.5553	0.1648
3.650	76.1478	12.5735	0.1651
3.655	76.1452	12.5364	0.1646
3.660	76.1412	12.5828	0.1653
3.665	76.1424	12.6064	0.1656
3.670	76.1060	12.6103	0.1657
3.675	76.0737	12.5782	0.1653
3.680	76.0686	12.6802	0.1667
3.685	76.1402	12.7058	0.1669
3.690	76.1277	12.7199	0.1671
3.695	76.1114	12.7125	0.1670
3.700	76.0531	12.7223	0.1673
3.705	76.0787	12.7841	0.1680
3.710	76.0682	12.7959	0.1682
3.715	76.0658	12.8501	0.1689
3.720	76.0595	12.8538	0.1690
3.725	76.1082	12.8609	0.1690
3.730	76.1169	12.8862	0.1693
3.735	76.1096	12.9169	0.1697
3.740	76.1123	12.8846	0.1693
3.745	76.0982	12.8604	0.1690
3.750	76.1384	12.8756	0.1691
3.755	76.1071	12.8786	0.1692
3.760	76.0735	12.8654	0.1691

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.765	76.0796	12.9422	0.1701
3.770	76.0365	12.9267	0.1700
3.775	76.0260	12.9543	0.1704
3.780	76.0710	12.9533	0.1703
3.785	76.0608	13.0349	0.1714
3.790	76.0329	13.0428	0.1715
3.795	75.9946	13.0979	0.1724
3.800	75.9992	13.0864	0.1722
3.805	76.0393	13.1153	0.1725
3.810	76.0205	13.1330	0.1728
3.815	76.0436	13.1458	0.1729
3.820	76.0020	13.1629	0.1732
3.825	76.0448	13.1951	0.1735
3.830	76.0353	13.2180	0.1738
3.835	76.1843	13.6353	0.1790
3.840	76.0197	13.2233	0.1739
3.845	76.0264	13.2004	0.1736
3.850	76.0134	13.2402	0.1742
3.855	75.9838	13.2036	0.1738
3.860	75.9791	13.2615	0.1745
3.865	75.9701	13.2503	0.1744
3.870	75.9612	13.2848	0.1749
3.875	75.9589	13.3257	0.1754
3.880	75.9803	13.3204	0.1753
3.885	75.9492	13.3772	0.1761
3.890	75.9327	13.3468	0.1758
3.895	75.9457	13.3642	0.1760
3.900	75.9396	13.3885	0.1763
3.905	75.9561	13.4003	0.1764
3.910	75.9220	13.4436	0.1771
3.915	75.9345	13.4570	0.1772
3.920	75.9239	13.4374	0.1770
3.925	75.8911	13.4361	0.1770
3.930	75.9106	13.4684	0.1774
3.935	75.8212	13.4892	0.1779
3.940	75.9184	13.5182	0.1781
3.945	75.8597	13.5035	0.1780
3.950	75.8780	13.5499	0.1786
3.955	75.8398	13.5780	0.1790
3.960	75.8451	13.5924	0.1792
3.965	75.8501	13.6083	0.1794
3.970	75.8366	13.6469	0.1800
3.975	75.8778	13.6196	0.1795
3.980	75.8295	13.6644	0.1802
3.985	75.8448	13.6704	0.1802
3.990	75.8004	13.7224	0.1810

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.995	75.8151	13.6894	0.1806
4.000	75.8138	13.7238	0.1810
4.005	75.7699	13.7045	0.1809
4.010	75.7387	13.7575	0.1816
4.015	75.7796	13.7834	0.1819
4.020	75.7307	13.7856	0.1820
4.025	75.7460	13.8072	0.1823
4.030	75.7360	13.8182	0.1825
4.035	75.7288	13.8266	0.1826
4.040	75.6961	13.8278	0.1827
4.045	75.6930	13.8553	0.1830
4.050	75.6984	13.8729	0.1833
4.055	75.6673	13.8995	0.1837
4.060	75.6841	13.8898	0.1835
4.065	75.6184	13.9254	0.1842
4.070	75.6272	13.9364	0.1843
4.075	75.6349	13.9443	0.1844
4.080	75.6362	13.9771	0.1848
4.085	75.5675	13.9983	0.1852
4.090	75.6063	14.0069	0.1853
4.095	75.6188	13.9767	0.1848
4.100	75.5730	14.0323	0.1857
4.105	75.5632	14.0035	0.1853
4.110	75.5663	14.0597	0.1861
4.115	75.5462	14.0762	0.1863
4.120	75.5573	14.0634	0.1861
4.125	75.4917	14.0749	0.1864
4.130	75.5300	14.0826	0.1865
4.135	75.5351	14.0860	0.1865
4.140	75.5265	14.1609	0.1875
4.145	75.4973	14.1506	0.1874
4.150	75.5023	14.1464	0.1874
4.155	75.4946	14.1796	0.1878
4.160	75.4571	14.2074	0.1883
4.165	75.4749	14.2372	0.1886
4.170	75.4990	14.2579	0.1888
4.175	75.4715	14.2523	0.1888
4.180	75.4254	14.2660	0.1891
4.185	75.4897	14.2408	0.1886
4.190	75.4118	14.2791	0.1893
4.195	75.4455	14.3262	0.1899
4.200	75.4454	14.3443	0.1901
4.205	75.4713	14.3527	0.1902
4.210	75.4510	14.3769	0.1905
4.215	75.3581	14.3839	0.1909
4.220	75.3942	14.4148	0.1912

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.225	75.3596	14.4174	0.1913
4.230	75.3882	14.4580	0.1918
4.235	75.3686	14.4828	0.1922
4.240	75.3736	14.4710	0.1920
4.245	75.3468	14.4571	0.1919
4.250	75.3429	14.5155	0.1927
4.255	75.3344	14.5216	0.1928
4.260	75.3757	14.5144	0.1926
4.265	75.3409	14.5929	0.1937
4.270	75.3328	14.5874	0.1936
4.275	75.3167	14.5763	0.1935
4.280	75.2798	14.6034	0.1940
4.285	75.3249	14.6342	0.1943
4.290	75.2800	14.6536	0.1947
4.295	75.2894	14.6660	0.1948
4.300	75.2729	14.7077	0.1954
4.305	75.2786	14.6967	0.1952
4.310	75.2392	14.7025	0.1954
4.315	75.2590	14.7272	0.1957
4.320	75.2646	14.7465	0.1959
4.325	75.2442	14.7722	0.1963
4.330	75.2617	14.7767	0.1963
4.335	75.2410	14.7999	0.1967
4.340	75.2362	14.8123	0.1969
4.345	75.2637	14.8423	0.1972
4.350	75.2176	14.8306	0.1972
4.355	75.2185	14.8635	0.1976
4.360	75.2177	14.8953	0.1980
4.365	75.2138	14.8529	0.1975
4.370	75.1675	14.8923	0.1981
4.375	75.2212	14.8782	0.1978
4.380	75.1794	14.9206	0.1985
4.385	75.1749	14.9564	0.1990
4.390	75.1634	14.9550	0.1990
4.395	75.1762	14.9556	0.1989
4.400	75.1682	14.9940	0.1995
4.405	75.1392	15.0077	0.1997
4.410	75.1518	15.0289	0.2000
4.415	75.1572	15.0405	0.2001
4.420	75.1056	15.0336	0.2002
4.425	75.0725	15.0086	0.1999
4.430	75.0941	15.1031	0.2011
4.435	75.1352	15.1092	0.2011
4.440	75.1211	15.0982	0.2010
4.445	75.1258	15.1686	0.2019
4.450	75.1151	15.1313	0.2014

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.455	75.0580	15.1920	0.2024
4.460	75.1637	15.1648	0.2018
4.465	75.1249	15.1308	0.2014
4.470	75.1326	15.2111	0.2025
4.475	75.1066	15.2261	0.2027
4.480	75.0775	15.2205	0.2027
4.485	75.0544	15.1769	0.2022
4.490	75.0943	15.2446	0.2030
4.495	75.0794	15.2881	0.2036
4.500	75.0695	15.2365	0.2030
4.505	75.0784	15.2556	0.2032
4.510	75.0145	15.2088	0.2027
4.515	75.0281	15.2592	0.2034
4.520	75.0174	15.3857	0.2051
4.525	74.9988	15.2927	0.2039
4.530	74.9626	15.3425	0.2047
4.535	74.9638	15.3338	0.2045
4.540	74.9820	15.4051	0.2055
4.545	75.0012	15.4317	0.2058
4.550	74.9435	15.4311	0.2059
4.555	74.9485	15.4046	0.2055
4.560	74.9719	15.4388	0.2059
4.565	74.9354	15.4247	0.2058
4.570	74.9372	15.4657	0.2064
4.575	74.8395	15.5115	0.2073
4.580	74.8985	15.4889	0.2068
4.585	74.8765	15.4725	0.2066
4.590	74.8599	15.5453	0.2077
4.595	74.8853	15.5432	0.2076
4.600	74.8776	15.5914	0.2082
4.605	74.9070	15.5639	0.2078
4.610	74.8675	15.6054	0.2084
4.615	74.8167	15.5281	0.2075
4.620	74.8384	15.6220	0.2087
4.625	74.7714	15.6575	0.2094
4.630	74.8266	15.6383	0.2090
4.635	74.7943	15.6409	0.2091
4.640	74.8105	15.6526	0.2092
4.645	74.7890	15.6626	0.2094
4.650	74.8040	15.6724	0.2095
4.655	74.7397	15.7188	0.2103
4.660	74.7399	15.7002	0.2101
4.665	74.7719	15.7345	0.2104
4.670	74.7042	15.7224	0.2105
4.675	74.7436	15.7771	0.2111
4.680	74.6949	15.7478	0.2108

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.685	74.7225	15.8399	0.2120
4.690	74.7033	15.7996	0.2115
4.695	74.6512	15.8011	0.2117
4.700	74.6846	15.8208	0.2118
4.705	74.7150	15.8296	0.2119
4.710	74.6595	15.8362	0.2121
4.715	74.6600	15.8512	0.2123
4.720	74.6800	15.8408	0.2121
4.725	74.6789	15.9059	0.2130
4.730	74.5873	15.8839	0.2130
4.735	74.5710	15.9555	0.2140
4.740	74.6044	16.0202	0.2147
4.745	74.6342	15.9652	0.2139
4.750	74.5734	15.9588	0.2140
4.755	74.5645	16.0092	0.2147
4.760	74.6137	15.9844	0.2142
4.765	74.5523	16.0080	0.2147
4.770	74.5398	16.0443	0.2152
4.775	74.6146	16.0074	0.2145
4.780	74.5019	16.0611	0.2156
4.785	74.4680	16.0599	0.2157
4.790	74.5124	16.0454	0.2153
4.795	74.5047	16.0715	0.2157
4.800	74.5240	16.1153	0.2162
4.805	74.5010	16.1276	0.2165
4.810	74.4502	16.0838	0.2160
4.815	74.4483	16.2118	0.2178
4.820	74.4350	16.1617	0.2171
4.825	74.4521	16.2032	0.2176
4.830	74.4667	16.2280	0.2179
4.835	74.4289	16.2245	0.2180
4.840	74.4317	16.2533	0.2184
4.845	74.4111	16.2375	0.2182
4.850	74.4681	16.2499	0.2182
4.855	74.4276	16.3054	0.2191
4.860	74.4400	16.3488	0.2196
4.865	74.3430	16.2462	0.2185
4.870	74.3761	16.3493	0.2198
4.875	74.4075	16.3646	0.2199
4.880	74.3968	16.3982	0.2204
4.885	74.3837	16.4305	0.2209
4.890	74.3973	16.4702	0.2214
4.895	74.3355	16.5014	0.2220
4.900	74.3447	16.4623	0.2214
4.905	74.3346	16.4998	0.2220
4.910	74.2992	16.4695	0.2217

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.915	74.3118	16.4865	0.2219
4.920	74.2828	16.5222	0.2224
4.925	74.2803	16.5142	0.2223
4.930	74.2939	16.5095	0.2222
4.935	74.2808	16.5163	0.2223
4.940	74.2798	16.5445	0.2227
4.945	74.2850	16.5509	0.2228
4.950	74.3003	16.5390	0.2226
4.955	74.2826	16.5553	0.2229
4.960	74.2978	16.5843	0.2232
4.965	74.3130	16.6015	0.2234
4.970	74.3245	16.6027	0.2234
4.975	74.3194	16.6019	0.2234
4.980	74.3336	16.6038	0.2234
4.985	74.3620	16.6542	0.2240
4.990	74.3550	16.6785	0.2243
4.995	74.3742	16.6818	0.2243
5.000	74.3763	16.7252	0.2249
5.005	74.3545	16.7536	0.2253
5.010	74.3680	16.7769	0.2256
5.015	74.3860	16.7833	0.2256
5.020	74.3992	16.8558	0.2266
5.025	74.3945	16.8887	0.2270
5.030	74.3739	16.9166	0.2275
5.035	74.3151	16.9919	0.2286
5.040	74.3045	17.0524	0.2295
5.045	74.2886	17.0455	0.2294
5.050	74.2533	17.0912	0.2302
5.055	74.1766	17.0748	0.2302
5.060	74.1435	17.1036	0.2307
5.065	74.0554	17.0740	0.2306
5.070	74.0333	17.0669	0.2305
5.075	73.9929	17.0586	0.2305
5.080	74.0264	17.0990	0.2310
5.085	73.9494	17.0701	0.2308
5.090	73.9570	16.9923	0.2298
5.095	74.0464	17.0873	0.2308
5.100	73.9635	17.0093	0.2300
5.105	73.9659	16.9735	0.2295
5.110	74.0411	17.0261	0.2300
5.115	74.0011	17.0608	0.2305
5.120	74.0414	17.0199	0.2299
5.125	74.0108	17.0691	0.2306
5.130	74.0536	17.0773	0.2306
5.135	74.0943	17.0571	0.2302
5.140	74.0390	17.1128	0.2311

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.145	74.0281	17.1851	0.2321
5.150	74.0662	17.1544	0.2316
5.155	73.9897	17.1956	0.2324
5.160	74.0251	17.1812	0.2321
5.165	73.9861	17.1986	0.2325
5.170	73.9810	17.2642	0.2334
5.175	73.9609	17.2655	0.2334
5.180	73.9661	17.2588	0.2333
5.185	73.9683	17.2625	0.2334
5.190	73.9687	17.2943	0.2338
5.195	73.9601	17.3156	0.2341
5.200	73.9391	17.3019	0.2340
5.205	73.8961	17.3061	0.2342
5.210	73.9098	17.3387	0.2346
5.215	73.9136	17.3598	0.2349
5.220	73.8848	17.3782	0.2352
5.225	73.8221	17.3381	0.2349
5.230	73.8347	17.3804	0.2354
5.235	73.8326	17.3855	0.2355
5.240	73.8180	17.4337	0.2362
5.245	73.8402	17.4673	0.2366
5.250	73.8216	17.4761	0.2367
5.255	73.7965	17.4974	0.2371
5.260	73.8085	17.4935	0.2370
5.265	73.7454	17.4995	0.2373
5.270	73.7300	17.5320	0.2378
5.275	73.7492	17.5267	0.2377
5.280	73.7695	17.5343	0.2377
5.285	73.7312	17.5479	0.2380
5.290	73.7399	17.5693	0.2383
5.295	73.6803	17.5754	0.2385
5.300	73.7188	17.5787	0.2385
5.305	73.6918	17.6416	0.2394
5.310	73.7147	17.6207	0.2390
5.315	73.6854	17.6315	0.2393
5.320	73.6399	17.6702	0.2400
5.325	73.6178	17.6317	0.2395
5.330	73.6404	17.6756	0.2400
5.335	73.6641	17.7002	0.2403
5.340	73.6057	17.6667	0.2400
5.345	73.5407	17.6766	0.2404
5.350	73.5784	17.7004	0.2406
5.355	73.5638	17.7244	0.2409
5.360	73.5809	17.7839	0.2417
5.365	73.5073	17.8237	0.2425
5.370	73.5577	17.8034	0.2420

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.375	73.4956	17.8284	0.2426
5.380	73.5044	17.8736	0.2432
5.385	73.5669	17.7716	0.2416
5.390	73.5252	17.8439	0.2427
5.395	73.5398	17.8955	0.2433
5.400	73.4742	17.9019	0.2436
5.405	73.4818	17.8746	0.2433
5.410	73.4874	17.9121	0.2437
5.415	73.4902	18.0004	0.2449
5.420	73.4591	17.9477	0.2443
5.425	73.4275	17.9509	0.2445
5.430	73.4153	18.0295	0.2456
5.435	73.4398	18.0032	0.2451
5.440	73.4424	18.0218	0.2454
5.445	73.4238	18.0405	0.2457
5.450	73.4196	18.0142	0.2454
5.455	73.3797	17.9985	0.2453
5.460	73.3694	18.0550	0.2461
5.465	73.3180	18.0488	0.2462
5.470	73.4087	18.1097	0.2467
5.475	73.3477	18.1358	0.2473
5.480	73.3866	18.1502	0.2473
5.485	73.3592	18.1851	0.2479
5.490	73.3955	18.1843	0.2478
5.495	73.3657	18.2057	0.2481
5.500	73.3477	18.2044	0.2482
5.505	73.3571	18.1595	0.2475
5.510	73.3161	18.2725	0.2492
5.515	73.3161	18.3022	0.2496
5.520	73.3206	18.3097	0.2497
5.525	73.2993	18.3147	0.2499
5.530	73.2929	18.3186	0.2499
5.535	73.2930	18.3644	0.2506
5.540	73.2906	18.3417	0.2503
5.545	73.3064	18.3237	0.2500
5.550	73.2741	18.3710	0.2507
5.555	73.2572	18.3911	0.2510
5.560	73.2783	18.4314	0.2515
5.565	73.2682	18.4246	0.2515
5.570	73.2437	18.4352	0.2517
5.575	73.2490	18.4683	0.2521
5.580	73.2534	18.4725	0.2522
5.585	73.2407	18.4875	0.2524
5.590	73.2338	18.5203	0.2529
5.595	73.2338	18.5263	0.2530
5.600	73.2229	18.5486	0.2533

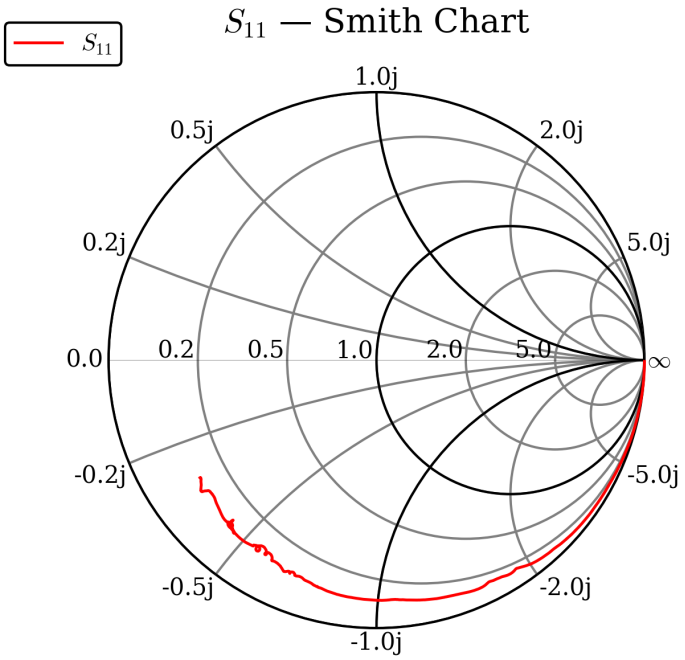
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.605	73.2106	18.5662	0.2536
5.610	73.2062	18.5608	0.2535
5.615	73.2454	18.5857	0.2537
5.620	73.1823	18.5880	0.2540
5.625	73.2100	18.5922	0.2540
5.630	73.1935	18.6091	0.2542
5.635	73.2115	18.6107	0.2542
5.640	73.1858	18.6565	0.2549
5.645	73.1646	18.6601	0.2550
5.650	73.1657	18.6513	0.2549
5.655	73.1795	18.6562	0.2549
5.660	73.1501	18.6480	0.2549
5.665	73.1457	18.6954	0.2556
5.670	73.1765	18.6885	0.2554
5.675	73.1253	18.7036	0.2558
5.680	73.1243	18.6918	0.2556
5.685	73.1102	18.6726	0.2554
5.690	73.0804	18.7152	0.2561
5.695	73.0455	18.7949	0.2573
5.700	73.0924	18.7504	0.2565
5.705	73.0184	18.7967	0.2574
5.710	73.1119	18.7715	0.2567
5.715	73.0411	18.8690	0.2583
5.720	73.0448	18.8940	0.2587
5.725	73.1048	18.8471	0.2578
5.730	73.0779	18.8568	0.2580
5.735	73.0839	18.8536	0.2580
5.740	73.0815	18.8195	0.2575
5.745	72.9156	18.8760	0.2589
5.750	72.9869	18.8654	0.2585
5.755	73.0155	18.8967	0.2588
5.760	72.9937	18.8943	0.2588
5.765	72.9067	18.9023	0.2593
5.770	73.0048	18.9384	0.2594
5.775	72.9117	18.9219	0.2595
5.780	72.9679	18.9663	0.2599
5.785	72.9156	18.9779	0.2603
5.790	72.9224	18.9948	0.2605
5.795	72.8773	19.0052	0.2608
5.800	72.8450	19.0004	0.2608
5.805	72.8656	18.9820	0.2605
5.810	72.8869	18.9847	0.2605
5.815	72.8837	19.0586	0.2615
5.820	72.8156	19.0251	0.2613
5.825	72.8740	19.0944	0.2620
5.830	72.8257	19.0792	0.2620

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.835	72.7727	19.0834	0.2622
5.840	72.7865	19.0747	0.2621
5.845	72.7367	19.0752	0.2622
5.850	72.7031	19.1097	0.2628
5.855	72.7572	19.1335	0.2630
5.860	72.7135	19.0976	0.2626
5.865	72.7249	19.1704	0.2636
5.870	72.6636	19.1661	0.2638
5.875	72.7001	19.1991	0.2641
5.880	72.6548	19.1675	0.2638
5.885	72.6604	19.2030	0.2643
5.890	72.6709	19.1790	0.2639
5.895	72.6312	19.2341	0.2648
5.900	72.5997	19.2310	0.2649
5.905	72.6074	19.2859	0.2656
5.910	72.6064	19.2682	0.2654
5.915	72.6020	19.2912	0.2657
5.920	72.5739	19.2895	0.2658
5.925	72.5474	19.3560	0.2668
5.930	72.5024	19.2922	0.2661
5.935	72.5180	19.3224	0.2664
5.940	72.5055	19.3773	0.2673
5.945	72.5422	19.3429	0.2666
5.950	72.5422	19.3701	0.2670
5.955	72.4879	19.3762	0.2673
5.960	72.5155	19.3970	0.2675
5.965	72.5560	19.4437	0.2680
5.970	72.4766	19.3756	0.2673
5.975	72.4068	19.4529	0.2687
5.980	72.4781	19.4607	0.2685
5.985	72.3630	19.4389	0.2686
5.990	72.4459	19.4754	0.2688
5.995	72.4045	19.5235	0.2696
6.000	72.4221	19.5024	0.2693

1.2 S_{11} — Smith Chart



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	1.0000	-0.0051	0.00
0.010	1.0000	-0.0102	0.00
0.015	0.9998	-0.0151	-0.00
0.020	0.9996	-0.0202	-0.00
0.025	0.9997	-0.0250	-0.00
0.030	0.9995	-0.0300	-0.00
0.035	0.9992	-0.0349	-0.00
0.040	0.9991	-0.0399	-0.00
0.045	0.9989	-0.0448	-0.00
0.050	0.9986	-0.0499	-0.00
0.055	0.9983	-0.0548	-0.00
0.060	0.9980	-0.0598	-0.00
0.065	0.9976	-0.0648	-0.00
0.070	0.9973	-0.0697	-0.00
0.075	0.9968	-0.0746	-0.00
0.080	0.9965	-0.0795	-0.00
0.085	0.9959	-0.0844	-0.00
0.090	0.9953	-0.0893	-0.01
0.095	0.9949	-0.0941	-0.01
0.100	0.9943	-0.0992	-0.01
0.105	0.9938	-0.1042	-0.01
0.110	0.9931	-0.1091	-0.01
0.115	0.9925	-0.1139	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9919	-0.1189	-0.01
0.125	0.9912	-0.1237	-0.01
0.130	0.9904	-0.1286	-0.01
0.135	0.9896	-0.1334	-0.01
0.140	0.9889	-0.1382	-0.01
0.145	0.9880	-0.1430	-0.01
0.150	0.9872	-0.1480	-0.02
0.155	0.9864	-0.1526	-0.02
0.160	0.9855	-0.1575	-0.02
0.165	0.9847	-0.1623	-0.02
0.170	0.9840	-0.1671	-0.02
0.175	0.9829	-0.1719	-0.02
0.180	0.9819	-0.1768	-0.02
0.185	0.9809	-0.1815	-0.02
0.190	0.9798	-0.1862	-0.02
0.195	0.9787	-0.1910	-0.02
0.200	0.9779	-0.1957	-0.02
0.205	0.9767	-0.2004	-0.03
0.210	0.9756	-0.2053	-0.03
0.215	0.9744	-0.2102	-0.03
0.220	0.9734	-0.2151	-0.03
0.225	0.9721	-0.2198	-0.03
0.230	0.9708	-0.2245	-0.03
0.235	0.9695	-0.2292	-0.03
0.240	0.9683	-0.2340	-0.03
0.245	0.9670	-0.2389	-0.03
0.250	0.9657	-0.2436	-0.04
0.255	0.9643	-0.2482	-0.04
0.260	0.9629	-0.2530	-0.04
0.265	0.9615	-0.2578	-0.04
0.270	0.9600	-0.2623	-0.04
0.275	0.9586	-0.2669	-0.04
0.280	0.9570	-0.2717	-0.05
0.285	0.9555	-0.2762	-0.05
0.290	0.9540	-0.2810	-0.05
0.295	0.9524	-0.2855	-0.05
0.300	0.9508	-0.2903	-0.05
0.305	0.9491	-0.2949	-0.05
0.310	0.9474	-0.2994	-0.06
0.315	0.9457	-0.3038	-0.06
0.320	0.9439	-0.3083	-0.06
0.325	0.9424	-0.3130	-0.06
0.330	0.9405	-0.3175	-0.06
0.335	0.9387	-0.3219	-0.07
0.340	0.9370	-0.3263	-0.07
0.345	0.9352	-0.3308	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9334	-0.3352	-0.07
0.355	0.9315	-0.3397	-0.07
0.360	0.9296	-0.3441	-0.08
0.365	0.9277	-0.3485	-0.08
0.370	0.9257	-0.3530	-0.08
0.375	0.9238	-0.3573	-0.08
0.380	0.9217	-0.3615	-0.09
0.385	0.9197	-0.3659	-0.09
0.390	0.9180	-0.3704	-0.09
0.395	0.9160	-0.3745	-0.09
0.400	0.9139	-0.3787	-0.09
0.405	0.9120	-0.3829	-0.10
0.410	0.9099	-0.3871	-0.10
0.415	0.9080	-0.3912	-0.10
0.420	0.9059	-0.3954	-0.10
0.425	0.9039	-0.3996	-0.10
0.430	0.9019	-0.4037	-0.10
0.435	0.8999	-0.4077	-0.11
0.440	0.8978	-0.4119	-0.11
0.445	0.8956	-0.4161	-0.11
0.450	0.8935	-0.4200	-0.11
0.455	0.8913	-0.4240	-0.11
0.460	0.8891	-0.4279	-0.12
0.465	0.8869	-0.4319	-0.12
0.470	0.8847	-0.4359	-0.12
0.475	0.8825	-0.4397	-0.12
0.480	0.8805	-0.4437	-0.12
0.485	0.8782	-0.4476	-0.13
0.490	0.8761	-0.4514	-0.13
0.495	0.8738	-0.4554	-0.13
0.500	0.8716	-0.4591	-0.13
0.505	0.8695	-0.4629	-0.13
0.510	0.8672	-0.4667	-0.13
0.515	0.8650	-0.4706	-0.13
0.520	0.8629	-0.4742	-0.13
0.525	0.8608	-0.4779	-0.14
0.530	0.8585	-0.4816	-0.14
0.535	0.8564	-0.4853	-0.14
0.540	0.8541	-0.4890	-0.14
0.545	0.8518	-0.4927	-0.14
0.550	0.8495	-0.4963	-0.14
0.555	0.8471	-0.5002	-0.14
0.560	0.8448	-0.5038	-0.14
0.565	0.8425	-0.5073	-0.14
0.570	0.8402	-0.5109	-0.15
0.575	0.8378	-0.5147	-0.15

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8355	-0.5181	-0.15
0.585	0.8331	-0.5217	-0.15
0.590	0.8308	-0.5253	-0.15
0.595	0.8284	-0.5287	-0.15
0.600	0.8260	-0.5322	-0.15
0.605	0.8235	-0.5356	-0.15
0.610	0.8211	-0.5389	-0.16
0.615	0.8188	-0.5424	-0.16
0.620	0.8163	-0.5458	-0.16
0.625	0.8139	-0.5492	-0.16
0.630	0.8116	-0.5526	-0.16
0.635	0.8091	-0.5560	-0.16
0.640	0.8068	-0.5594	-0.16
0.645	0.8043	-0.5627	-0.16
0.650	0.8018	-0.5661	-0.16
0.655	0.7995	-0.5694	-0.16
0.660	0.7970	-0.5727	-0.16
0.665	0.7945	-0.5760	-0.16
0.670	0.7919	-0.5792	-0.17
0.675	0.7894	-0.5825	-0.17
0.680	0.7868	-0.5857	-0.17
0.685	0.7843	-0.5887	-0.17
0.690	0.7816	-0.5919	-0.17
0.695	0.7790	-0.5950	-0.17
0.700	0.7762	-0.5982	-0.18
0.705	0.7735	-0.6015	-0.18
0.710	0.7707	-0.6047	-0.18
0.715	0.7682	-0.6079	-0.18
0.720	0.7654	-0.6109	-0.18
0.725	0.7626	-0.6140	-0.18
0.730	0.7598	-0.6171	-0.19
0.735	0.7571	-0.6203	-0.19
0.740	0.7544	-0.6232	-0.19
0.745	0.7515	-0.6262	-0.19
0.750	0.7488	-0.6293	-0.19
0.755	0.7458	-0.6322	-0.20
0.760	0.7429	-0.6350	-0.20
0.765	0.7400	-0.6381	-0.20
0.770	0.7372	-0.6412	-0.20
0.775	0.7341	-0.6441	-0.21
0.780	0.7311	-0.6472	-0.21
0.785	0.7280	-0.6502	-0.21
0.790	0.7249	-0.6530	-0.21
0.795	0.7219	-0.6559	-0.22
0.800	0.7187	-0.6588	-0.22
0.805	0.7156	-0.6617	-0.22

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.810	0.7124	-0.6644	-0.23
0.815	0.7093	-0.6673	-0.23
0.820	0.7061	-0.6699	-0.23
0.825	0.7030	-0.6725	-0.24
0.830	0.6999	-0.6753	-0.24
0.835	0.6965	-0.6781	-0.25
0.840	0.6933	-0.6809	-0.25
0.845	0.6901	-0.6836	-0.25
0.850	0.6867	-0.6862	-0.26
0.855	0.6832	-0.6890	-0.26
0.860	0.6797	-0.6914	-0.27
0.865	0.6763	-0.6939	-0.27
0.870	0.6730	-0.6961	-0.28
0.875	0.6697	-0.6985	-0.29
0.880	0.6665	-0.7006	-0.29
0.885	0.6634	-0.7029	-0.30
0.890	0.6604	-0.7053	-0.30
0.895	0.6573	-0.7078	-0.30
0.900	0.6544	-0.7104	-0.30
0.905	0.6513	-0.7133	-0.30
0.910	0.6479	-0.7158	-0.30
0.915	0.6446	-0.7187	-0.31
0.920	0.6411	-0.7212	-0.31
0.925	0.6376	-0.7239	-0.31
0.930	0.6339	-0.7266	-0.32
0.935	0.6303	-0.7291	-0.32
0.940	0.6267	-0.7316	-0.32
0.945	0.6231	-0.7340	-0.33
0.950	0.6193	-0.7365	-0.33
0.955	0.6155	-0.7390	-0.34
0.960	0.6118	-0.7412	-0.34
0.965	0.6081	-0.7436	-0.35
0.970	0.6044	-0.7459	-0.35
0.975	0.6007	-0.7480	-0.36
0.980	0.5969	-0.7502	-0.37
0.985	0.5930	-0.7523	-0.37
0.990	0.5892	-0.7544	-0.38
0.995	0.5852	-0.7564	-0.39
1.000	0.5813	-0.7583	-0.40
1.005	0.5773	-0.7602	-0.40
1.010	0.5732	-0.7618	-0.41
1.015	0.5692	-0.7632	-0.43
1.020	0.5651	-0.7647	-0.44
1.025	0.5613	-0.7661	-0.45
1.030	0.5574	-0.7674	-0.46
1.035	0.5537	-0.7685	-0.47

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.5501	-0.7695	-0.48
1.045	0.5467	-0.7705	-0.49
1.050	0.5434	-0.7712	-0.51
1.055	0.5402	-0.7721	-0.52
1.060	0.5373	-0.7732	-0.52
1.065	0.5342	-0.7740	-0.53
1.070	0.5312	-0.7750	-0.54
1.075	0.5284	-0.7757	-0.55
1.080	0.5258	-0.7763	-0.56
1.085	0.5236	-0.7769	-0.57
1.090	0.5218	-0.7778	-0.57
1.095	0.5200	-0.7791	-0.57
1.100	0.5183	-0.7809	-0.56
1.105	0.5166	-0.7831	-0.55
1.110	0.5145	-0.7855	-0.55
1.115	0.5120	-0.7879	-0.54
1.120	0.5093	-0.7906	-0.53
1.125	0.5061	-0.7930	-0.53
1.130	0.5029	-0.7954	-0.53
1.135	0.4993	-0.7976	-0.53
1.140	0.4958	-0.7994	-0.53
1.145	0.4922	-0.8011	-0.53
1.150	0.4887	-0.8026	-0.54
1.155	0.4850	-0.8038	-0.55
1.160	0.4816	-0.8048	-0.56
1.165	0.4784	-0.8057	-0.57
1.170	0.4753	-0.8065	-0.57
1.175	0.4725	-0.8072	-0.58
1.180	0.4698	-0.8081	-0.59
1.185	0.4675	-0.8088	-0.59
1.190	0.4650	-0.8099	-0.59
1.195	0.4628	-0.8110	-0.60
1.200	0.4603	-0.8121	-0.60
1.205	0.4576	-0.8130	-0.60
1.210	0.4551	-0.8139	-0.61
1.215	0.4525	-0.8150	-0.61
1.220	0.4501	-0.8159	-0.61
1.225	0.4479	-0.8169	-0.62
1.230	0.4455	-0.8181	-0.62
1.235	0.4431	-0.8191	-0.62
1.240	0.4406	-0.8205	-0.62
1.245	0.4380	-0.8216	-0.62
1.250	0.4354	-0.8226	-0.62
1.255	0.4329	-0.8233	-0.63
1.260	0.4305	-0.8240	-0.63
1.265	0.4280	-0.8248	-0.64

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.4260	-0.8252	-0.64
1.275	0.4238	-0.8259	-0.65
1.280	0.4219	-0.8266	-0.65
1.285	0.4202	-0.8273	-0.65
1.290	0.4185	-0.8283	-0.65
1.295	0.4170	-0.8297	-0.64
1.300	0.4153	-0.8312	-0.64
1.305	0.4137	-0.8328	-0.63
1.310	0.4121	-0.8344	-0.62
1.315	0.4101	-0.8363	-0.62
1.320	0.4080	-0.8381	-0.61
1.325	0.4058	-0.8399	-0.60
1.330	0.4033	-0.8416	-0.60
1.335	0.4008	-0.8434	-0.59
1.340	0.3983	-0.8449	-0.59
1.345	0.3954	-0.8467	-0.59
1.350	0.3926	-0.8481	-0.59
1.355	0.3897	-0.8494	-0.59
1.360	0.3867	-0.8510	-0.59
1.365	0.3838	-0.8525	-0.58
1.370	0.3808	-0.8537	-0.59
1.375	0.3778	-0.8547	-0.59
1.380	0.3747	-0.8559	-0.59
1.385	0.3716	-0.8570	-0.59
1.390	0.3686	-0.8581	-0.59
1.395	0.3654	-0.8593	-0.60
1.400	0.3623	-0.8603	-0.60
1.405	0.3589	-0.8612	-0.60
1.410	0.3557	-0.8621	-0.61
1.415	0.3524	-0.8630	-0.61
1.420	0.3491	-0.8641	-0.61
1.425	0.3458	-0.8648	-0.62
1.430	0.3425	-0.8657	-0.62
1.435	0.3392	-0.8663	-0.63
1.440	0.3359	-0.8672	-0.63
1.445	0.3326	-0.8678	-0.64
1.450	0.3294	-0.8684	-0.64
1.455	0.3264	-0.8691	-0.65
1.460	0.3230	-0.8698	-0.65
1.465	0.3199	-0.8704	-0.66
1.470	0.3168	-0.8709	-0.66
1.475	0.3137	-0.8717	-0.66
1.480	0.3107	-0.8724	-0.67
1.485	0.3073	-0.8731	-0.67
1.490	0.3042	-0.8739	-0.67
1.495	0.3011	-0.8746	-0.68

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.2979	-0.8753	-0.68
1.505	0.2944	-0.8759	-0.69
1.510	0.2912	-0.8767	-0.69
1.515	0.2879	-0.8775	-0.69
1.520	0.2845	-0.8783	-0.69
1.525	0.2812	-0.8791	-0.70
1.530	0.2780	-0.8799	-0.70
1.535	0.2744	-0.8805	-0.70
1.540	0.2709	-0.8814	-0.70
1.545	0.2672	-0.8821	-0.71
1.550	0.2639	-0.8828	-0.71
1.555	0.2603	-0.8835	-0.71
1.560	0.2565	-0.8840	-0.72
1.565	0.2529	-0.8849	-0.72
1.570	0.2492	-0.8857	-0.72
1.575	0.2455	-0.8864	-0.73
1.580	0.2418	-0.8869	-0.73
1.585	0.2383	-0.8876	-0.73
1.590	0.2345	-0.8883	-0.74
1.595	0.2309	-0.8890	-0.74
1.600	0.2273	-0.8895	-0.74
1.605	0.2237	-0.8902	-0.74
1.610	0.2202	-0.8904	-0.75
1.615	0.2165	-0.8908	-0.75
1.620	0.2126	-0.8912	-0.76
1.625	0.2092	-0.8918	-0.76
1.630	0.2054	-0.8921	-0.77
1.635	0.2018	-0.8923	-0.77
1.640	0.1981	-0.8927	-0.78
1.645	0.1944	-0.8929	-0.78
1.650	0.1906	-0.8932	-0.79
1.655	0.1872	-0.8938	-0.79
1.660	0.1836	-0.8940	-0.79
1.665	0.1801	-0.8945	-0.80
1.670	0.1765	-0.8945	-0.80
1.675	0.1730	-0.8947	-0.81
1.680	0.1695	-0.8950	-0.81
1.685	0.1660	-0.8950	-0.82
1.690	0.1626	-0.8952	-0.82
1.695	0.1594	-0.8953	-0.82
1.700	0.1561	-0.8952	-0.83
1.705	0.1526	-0.8954	-0.84
1.710	0.1491	-0.8954	-0.84
1.715	0.1458	-0.8952	-0.85
1.720	0.1423	-0.8953	-0.85
1.725	0.1393	-0.8953	-0.86

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.1358	-0.8952	-0.86
1.735	0.1328	-0.8953	-0.87
1.740	0.1298	-0.8951	-0.87
1.745	0.1267	-0.8951	-0.88
1.750	0.1239	-0.8951	-0.88
1.755	0.1212	-0.8951	-0.88
1.760	0.1185	-0.8950	-0.89
1.765	0.1158	-0.8952	-0.89
1.770	0.1134	-0.8951	-0.89
1.775	0.1109	-0.8950	-0.90
1.780	0.1084	-0.8948	-0.90
1.785	0.1060	-0.8945	-0.91
1.790	0.1033	-0.8944	-0.91
1.795	0.1007	-0.8942	-0.92
1.800	0.0982	-0.8940	-0.92
1.805	0.0956	-0.8942	-0.92
1.810	0.0934	-0.8942	-0.92
1.815	0.0912	-0.8941	-0.93
1.820	0.0888	-0.8942	-0.93
1.825	0.0868	-0.8942	-0.93
1.830	0.0847	-0.8941	-0.93
1.835	0.0824	-0.8938	-0.94
1.840	0.0803	-0.8938	-0.94
1.845	0.0783	-0.8932	-0.95
1.850	0.0762	-0.8934	-0.95
1.855	0.0742	-0.8931	-0.95
1.860	0.0720	-0.8930	-0.95
1.865	0.0700	-0.8929	-0.96
1.870	0.0680	-0.8930	-0.96
1.875	0.0660	-0.8931	-0.96
1.880	0.0641	-0.8934	-0.96
1.885	0.0625	-0.8935	-0.96
1.890	0.0609	-0.8937	-0.96
1.895	0.0591	-0.8936	-0.96
1.900	0.0572	-0.8936	-0.96
1.905	0.0555	-0.8937	-0.96
1.910	0.0537	-0.8939	-0.96
1.915	0.0516	-0.8941	-0.96
1.920	0.0498	-0.8944	-0.96
1.925	0.0482	-0.8948	-0.95
1.930	0.0465	-0.8947	-0.95
1.935	0.0448	-0.8951	-0.95
1.940	0.0429	-0.8951	-0.95
1.945	0.0412	-0.8953	-0.95
1.950	0.0395	-0.8955	-0.95
1.955	0.0379	-0.8956	-0.95

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.0362	-0.8957	-0.95
1.965	0.0344	-0.8957	-0.95
1.970	0.0328	-0.8959	-0.95
1.975	0.0311	-0.8960	-0.95
1.980	0.0294	-0.8960	-0.95
1.985	0.0278	-0.8961	-0.95
1.990	0.0260	-0.8960	-0.95
1.995	0.0242	-0.8963	-0.95
2.000	0.0223	-0.8964	-0.95
2.005	0.0204	-0.8964	-0.95
2.010	0.0186	-0.8965	-0.95
2.015	0.0168	-0.8963	-0.95
2.020	0.0148	-0.8964	-0.95
2.025	0.0129	-0.8965	-0.95
2.030	0.0111	-0.8965	-0.95
2.035	0.0090	-0.8964	-0.95
2.040	0.0071	-0.8964	-0.95
2.045	0.0051	-0.8962	-0.95
2.050	0.0031	-0.8962	-0.95
2.055	0.0011	-0.8961	-0.95
2.060	-0.0011	-0.8959	-0.95
2.065	-0.0031	-0.8959	-0.95
2.070	-0.0054	-0.8957	-0.96
2.075	-0.0075	-0.8956	-0.96
2.080	-0.0097	-0.8954	-0.96
2.085	-0.0121	-0.8951	-0.96
2.090	-0.0144	-0.8949	-0.96
2.095	-0.0169	-0.8948	-0.96
2.100	-0.0192	-0.8945	-0.97
2.105	-0.0217	-0.8945	-0.97
2.110	-0.0242	-0.8942	-0.97
2.115	-0.0268	-0.8943	-0.97
2.120	-0.0295	-0.8941	-0.97
2.125	-0.0321	-0.8939	-0.97
2.130	-0.0347	-0.8938	-0.97
2.135	-0.0375	-0.8934	-0.97
2.140	-0.0403	-0.8933	-0.97
2.145	-0.0429	-0.8929	-0.97
2.150	-0.0456	-0.8927	-0.97
2.155	-0.0486	-0.8924	-0.98
2.160	-0.0512	-0.8919	-0.98
2.165	-0.0540	-0.8917	-0.98
2.170	-0.0569	-0.8913	-0.98
2.175	-0.0597	-0.8909	-0.98
2.180	-0.0625	-0.8906	-0.98
2.185	-0.0653	-0.8901	-0.99

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.190	-0.0684	-0.8898	-0.99
2.195	-0.0712	-0.8892	-0.99
2.200	-0.0743	-0.8890	-0.99
2.205	-0.0771	-0.8884	-0.99
2.210	-0.0802	-0.8880	-1.00
2.215	-0.0829	-0.8875	-1.00
2.220	-0.0859	-0.8870	-1.00
2.225	-0.0890	-0.8866	-1.00
2.230	-0.0916	-0.8859	-1.01
2.235	-0.0948	-0.8855	-1.01
2.240	-0.0977	-0.8850	-1.01
2.245	-0.1007	-0.8845	-1.01
2.250	-0.1038	-0.8840	-1.01
2.255	-0.1068	-0.8838	-1.01
2.260	-0.1095	-0.8832	-1.01
2.265	-0.1124	-0.8825	-1.02
2.270	-0.1153	-0.8818	-1.02
2.275	-0.1182	-0.8812	-1.02
2.280	-0.1209	-0.8806	-1.02
2.285	-0.1236	-0.8800	-1.03
2.290	-0.1265	-0.8794	-1.03
2.295	-0.1291	-0.8787	-1.03
2.300	-0.1320	-0.8779	-1.03
2.305	-0.1347	-0.8773	-1.04
2.310	-0.1373	-0.8765	-1.04
2.315	-0.1400	-0.8758	-1.04
2.320	-0.1426	-0.8751	-1.05
2.325	-0.1453	-0.8745	-1.05
2.330	-0.1479	-0.8739	-1.05
2.335	-0.1504	-0.8731	-1.05
2.340	-0.1530	-0.8724	-1.05
2.345	-0.1554	-0.8716	-1.06
2.350	-0.1579	-0.8707	-1.06
2.355	-0.1602	-0.8699	-1.07
2.360	-0.1624	-0.8692	-1.07
2.365	-0.1646	-0.8685	-1.07
2.370	-0.1667	-0.8678	-1.07
2.375	-0.1688	-0.8671	-1.08
2.380	-0.1711	-0.8663	-1.08
2.385	-0.1729	-0.8657	-1.08
2.390	-0.1750	-0.8651	-1.08
2.395	-0.1772	-0.8644	-1.09
2.400	-0.1791	-0.8637	-1.09
2.405	-0.1808	-0.8629	-1.09
2.410	-0.1827	-0.8622	-1.10
2.415	-0.1844	-0.8616	-1.10

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.420	-0.1861	-0.8610	-1.10
2.425	-0.1877	-0.8600	-1.11
2.430	-0.1892	-0.8595	-1.11
2.435	-0.1907	-0.8588	-1.11
2.440	-0.1922	-0.8580	-1.12
2.445	-0.1938	-0.8575	-1.12
2.450	-0.1950	-0.8567	-1.12
2.455	-0.1961	-0.8561	-1.13
2.460	-0.1974	-0.8554	-1.13
2.465	-0.1986	-0.8549	-1.13
2.470	-0.1998	-0.8541	-1.14
2.475	-0.2010	-0.8537	-1.14
2.480	-0.2021	-0.8531	-1.14
2.485	-0.2033	-0.8525	-1.15
2.490	-0.2044	-0.8521	-1.15
2.495	-0.2056	-0.8515	-1.15
2.500	-0.2067	-0.8511	-1.15
2.505	-0.2077	-0.8507	-1.15
2.510	-0.2088	-0.8504	-1.15
2.515	-0.2098	-0.8501	-1.15
2.520	-0.2110	-0.8498	-1.15
2.525	-0.2119	-0.8495	-1.15
2.530	-0.2130	-0.8492	-1.16
2.535	-0.2140	-0.8490	-1.15
2.540	-0.2150	-0.8489	-1.15
2.545	-0.2160	-0.8485	-1.15
2.550	-0.2168	-0.8483	-1.15
2.555	-0.2174	-0.8478	-1.16
2.560	-0.2183	-0.8473	-1.16
2.565	-0.2187	-0.8468	-1.16
2.570	-0.2192	-0.8464	-1.17
2.575	-0.2193	-0.8459	-1.17
2.580	-0.2197	-0.8454	-1.18
2.585	-0.2201	-0.8455	-1.17
2.590	-0.2205	-0.8452	-1.17
2.595	-0.2210	-0.8455	-1.17
2.600	-0.2215	-0.8451	-1.17
2.605	-0.2219	-0.8452	-1.17
2.610	-0.2224	-0.8450	-1.17
2.615	-0.2230	-0.8448	-1.17
2.620	-0.2238	-0.8446	-1.17
2.625	-0.2241	-0.8445	-1.17
2.630	-0.2247	-0.8442	-1.17
2.635	-0.2252	-0.8440	-1.17
2.640	-0.2260	-0.8436	-1.18
2.645	-0.2266	-0.8434	-1.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	-0.2272	-0.8430	-1.18
2.655	-0.2276	-0.8425	-1.18
2.660	-0.2285	-0.8422	-1.18
2.665	-0.2291	-0.8419	-1.18
2.670	-0.2296	-0.8414	-1.19
2.675	-0.2304	-0.8412	-1.19
2.680	-0.2308	-0.8410	-1.19
2.685	-0.2312	-0.8409	-1.19
2.690	-0.2318	-0.8407	-1.19
2.695	-0.2322	-0.8405	-1.19
2.700	-0.2326	-0.8405	-1.19
2.705	-0.2330	-0.8402	-1.19
2.710	-0.2335	-0.8401	-1.19
2.715	-0.2341	-0.8397	-1.19
2.720	-0.2348	-0.8394	-1.19
2.725	-0.2351	-0.8388	-1.20
2.730	-0.2359	-0.8381	-1.20
2.735	-0.2367	-0.8375	-1.21
2.740	-0.2376	-0.8366	-1.21
2.745	-0.2385	-0.8362	-1.21
2.750	-0.2396	-0.8352	-1.22
2.755	-0.2406	-0.8347	-1.22
2.760	-0.2417	-0.8337	-1.23
2.765	-0.2429	-0.8329	-1.23
2.770	-0.2441	-0.8322	-1.24
2.775	-0.2454	-0.8313	-1.24
2.780	-0.2467	-0.8307	-1.24
2.785	-0.2481	-0.8299	-1.25
2.790	-0.2494	-0.8292	-1.25
2.795	-0.2509	-0.8285	-1.25
2.800	-0.2523	-0.8278	-1.26
2.805	-0.2536	-0.8273	-1.26
2.810	-0.2552	-0.8266	-1.26
2.815	-0.2565	-0.8260	-1.26
2.820	-0.2578	-0.8254	-1.26
2.825	-0.2592	-0.8247	-1.27
2.830	-0.2604	-0.8242	-1.27
2.835	-0.2620	-0.8236	-1.27
2.840	-0.2634	-0.8231	-1.27
2.845	-0.2648	-0.8225	-1.27
2.850	-0.2660	-0.8217	-1.27
2.855	-0.2671	-0.8211	-1.28
2.860	-0.2685	-0.8204	-1.28
2.865	-0.2696	-0.8198	-1.28
2.870	-0.2706	-0.8190	-1.28
2.875	-0.2716	-0.8181	-1.29

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	-0.2726	-0.8171	-1.30
2.885	-0.2734	-0.8160	-1.30
2.890	-0.2749	-0.8148	-1.31
2.895	-0.2762	-0.8133	-1.32
2.900	-0.2777	-0.8122	-1.33
2.905	-0.2794	-0.8112	-1.33
2.910	-0.2811	-0.8101	-1.34
2.915	-0.2830	-0.8092	-1.34
2.920	-0.2846	-0.8085	-1.34
2.925	-0.2863	-0.8079	-1.34
2.930	-0.2879	-0.8069	-1.34
2.935	-0.2895	-0.8067	-1.34
2.940	-0.2908	-0.8064	-1.34
2.945	-0.2923	-0.8062	-1.33
2.950	-0.2937	-0.8062	-1.33
2.955	-0.2948	-0.8060	-1.33
2.960	-0.2959	-0.8057	-1.33
2.965	-0.2967	-0.8056	-1.32
2.970	-0.2975	-0.8055	-1.32
2.975	-0.2983	-0.8057	-1.32
2.980	-0.2991	-0.8057	-1.32
2.985	-0.2995	-0.8055	-1.32
2.990	-0.3000	-0.8053	-1.32
2.995	-0.3003	-0.8051	-1.32
3.000	-0.3007	-0.8046	-1.32
3.005	-0.3043	-0.8034	-1.32
3.010	-0.3046	-0.8033	-1.32
3.015	-0.3049	-0.8033	-1.32
3.020	-0.3051	-0.8031	-1.32
3.025	-0.3053	-0.8032	-1.32
3.030	-0.3050	-0.8030	-1.32
3.035	-0.3051	-0.8025	-1.32
3.040	-0.3052	-0.8017	-1.33
3.045	-0.3053	-0.8009	-1.34
3.050	-0.3048	-0.7999	-1.35
3.055	-0.3054	-0.7994	-1.35
3.060	-0.3054	-0.7981	-1.37
3.065	-0.3058	-0.7970	-1.37
3.070	-0.3068	-0.7959	-1.38
3.075	-0.3073	-0.7949	-1.39
3.080	-0.3080	-0.7938	-1.40
3.085	-0.3091	-0.7928	-1.40
3.090	-0.3102	-0.7919	-1.41
3.095	-0.3113	-0.7914	-1.41
3.100	-0.3128	-0.7910	-1.41
3.105	-0.3139	-0.7908	-1.40

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	-0.3152	-0.7906	-1.40
3.115	-0.3169	-0.7905	-1.39
3.120	-0.3180	-0.7907	-1.39
3.125	-0.3195	-0.7909	-1.38
3.130	-0.3203	-0.7915	-1.37
3.135	-0.3213	-0.7920	-1.36
3.140	-0.3218	-0.7927	-1.36
3.145	-0.3226	-0.7933	-1.35
3.150	-0.3232	-0.7938	-1.34
3.155	-0.3234	-0.7943	-1.33
3.160	-0.3237	-0.7946	-1.33
3.165	-0.3238	-0.7951	-1.32
3.170	-0.3239	-0.7955	-1.32
3.175	-0.3238	-0.7954	-1.32
3.180	-0.3237	-0.7957	-1.32
3.185	-0.3236	-0.7955	-1.32
3.190	-0.3233	-0.7948	-1.33
3.195	-0.3234	-0.7941	-1.34
3.200	-0.3233	-0.7930	-1.35
3.205	-0.3236	-0.7923	-1.35
3.210	-0.3240	-0.7910	-1.36
3.215	-0.3243	-0.7899	-1.37
3.220	-0.3251	-0.7887	-1.38
3.225	-0.3268	-0.7881	-1.38
3.230	-0.3281	-0.7877	-1.38
3.235	-0.3294	-0.7877	-1.37
3.240	-0.3305	-0.7877	-1.37
3.245	-0.3316	-0.7879	-1.36
3.250	-0.3330	-0.7880	-1.36
3.255	-0.3342	-0.7882	-1.35
3.260	-0.3353	-0.7887	-1.34
3.265	-0.3364	-0.7889	-1.33
3.270	-0.3371	-0.7888	-1.33
3.275	-0.3383	-0.7891	-1.32
3.280	-0.3392	-0.7892	-1.32
3.285	-0.3405	-0.7891	-1.32
3.290	-0.3412	-0.7892	-1.31
3.295	-0.3421	-0.7892	-1.31
3.300	-0.3429	-0.7893	-1.30
3.305	-0.3434	-0.7891	-1.30
3.310	-0.3442	-0.7889	-1.30
3.315	-0.3448	-0.7887	-1.30
3.320	-0.3454	-0.7886	-1.30
3.325	-0.3460	-0.7885	-1.30
3.330	-0.3465	-0.7874	-1.31
3.335	-0.3474	-0.7869	-1.31

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.340	-0.3477	-0.7866	-1.31
3.345	-0.3484	-0.7852	-1.32
3.350	-0.3488	-0.7844	-1.33
3.355	-0.3496	-0.7834	-1.33
3.360	-0.3502	-0.7821	-1.34
3.365	-0.3510	-0.7812	-1.35
3.370	-0.3518	-0.7798	-1.36
3.375	-0.3522	-0.7784	-1.37
3.380	-0.3531	-0.7769	-1.38
3.385	-0.3542	-0.7756	-1.38
3.390	-0.3553	-0.7739	-1.40
3.395	-0.3566	-0.7723	-1.41
3.400	-0.3577	-0.7705	-1.42
3.405	-0.3585	-0.7684	-1.43
3.410	-0.3602	-0.7669	-1.44
3.415	-0.3618	-0.7652	-1.45
3.420	-0.3633	-0.7639	-1.45
3.425	-0.3649	-0.7625	-1.46
3.430	-0.3666	-0.7608	-1.47
3.435	-0.3683	-0.7593	-1.47
3.440	-0.3699	-0.7578	-1.48
3.445	-0.3719	-0.7564	-1.48
3.450	-0.3739	-0.7550	-1.49
3.455	-0.3759	-0.7536	-1.49
3.460	-0.3779	-0.7526	-1.49
3.465	-0.3794	-0.7518	-1.49
3.470	-0.3812	-0.7510	-1.49
3.475	-0.3830	-0.7503	-1.49
3.480	-0.3845	-0.7497	-1.49
3.485	-0.3863	-0.7492	-1.48
3.490	-0.3879	-0.7495	-1.47
3.495	-0.3889	-0.7491	-1.47
3.500	-0.3897	-0.7491	-1.47
3.505	-0.3903	-0.7489	-1.47
3.510	-0.3905	-0.7483	-1.47
3.515	-0.3908	-0.7478	-1.48
3.520	-0.3906	-0.7468	-1.49
3.525	-0.3914	-0.7457	-1.49
3.530	-0.3904	-0.7453	-1.50
3.535	-0.3896	-0.7431	-1.52
3.540	-0.3893	-0.7406	-1.55
3.545	-0.3893	-0.7385	-1.57
3.550	-0.3897	-0.7350	-1.60
3.555	-0.3906	-0.7319	-1.62
3.560	-0.3921	-0.7285	-1.65
3.565	-0.3935	-0.7258	-1.66

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	-0.3968	-0.7229	-1.68
3.575	-0.3999	-0.7205	-1.68
3.580	-0.4035	-0.7181	-1.68
3.585	-0.4072	-0.7175	-1.67
3.590	-0.4106	-0.7175	-1.65
3.595	-0.4137	-0.7188	-1.63
3.600	-0.4158	-0.7206	-1.60
3.605	-0.4168	-0.7228	-1.57
3.610	-0.4167	-0.7243	-1.56
3.615	-0.4163	-0.7254	-1.55
3.620	-0.4151	-0.7258	-1.55
3.625	-0.4131	-0.7263	-1.56
3.630	-0.4123	-0.7256	-1.57
3.635	-0.4108	-0.7244	-1.59
3.640	-0.4100	-0.7229	-1.61
3.645	-0.4097	-0.7208	-1.63
3.650	-0.4096	-0.7189	-1.65
3.655	-0.4101	-0.7173	-1.66
3.660	-0.4105	-0.7156	-1.67
3.665	-0.4110	-0.7140	-1.68
3.670	-0.4114	-0.7122	-1.70
3.675	-0.4116	-0.7104	-1.71
3.680	-0.4121	-0.7083	-1.73
3.685	-0.4133	-0.7063	-1.74
3.690	-0.4144	-0.7042	-1.76
3.695	-0.4161	-0.7021	-1.77
3.700	-0.4175	-0.7000	-1.78
3.705	-0.4201	-0.6974	-1.79
3.710	-0.4231	-0.6953	-1.79
3.715	-0.4266	-0.6938	-1.78
3.720	-0.4307	-0.6925	-1.77
3.725	-0.4354	-0.6930	-1.74
3.730	-0.4409	-0.6928	-1.71
3.735	-0.4444	-0.6951	-1.67
3.740	-0.4479	-0.6982	-1.62
3.745	-0.4492	-0.7015	-1.59
3.750	-0.4502	-0.7052	-1.55
3.755	-0.4494	-0.7085	-1.52
3.760	-0.4476	-0.7113	-1.51
3.765	-0.4447	-0.7124	-1.52
3.770	-0.4412	-0.7128	-1.53
3.775	-0.4382	-0.7109	-1.56
3.780	-0.4362	-0.7081	-1.60
3.785	-0.4354	-0.7040	-1.64
3.790	-0.4361	-0.7001	-1.67
3.795	-0.4384	-0.6970	-1.69

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	-0.4413	-0.6948	-1.69
3.805	-0.4439	-0.6930	-1.69
3.810	-0.4470	-0.6920	-1.68
3.815	-0.4499	-0.6911	-1.67
3.820	-0.4523	-0.6910	-1.66
3.825	-0.4550	-0.6905	-1.65
3.830	-0.4574	-0.6911	-1.63
3.835	-0.4578	-0.6899	-1.64
3.840	-0.4597	-0.6935	-1.60
3.845	-0.4611	-0.6935	-1.59
3.850	-0.4622	-0.6928	-1.59
3.855	-0.4634	-0.6925	-1.58
3.860	-0.4648	-0.6917	-1.58
3.865	-0.4664	-0.6910	-1.58
3.870	-0.4678	-0.6902	-1.58
3.875	-0.4693	-0.6894	-1.58
3.880	-0.4711	-0.6888	-1.57
3.885	-0.4728	-0.6879	-1.57
3.890	-0.4744	-0.6870	-1.57
3.895	-0.4761	-0.6861	-1.56
3.900	-0.4780	-0.6852	-1.56
3.905	-0.4800	-0.6842	-1.56
3.910	-0.4815	-0.6833	-1.56
3.915	-0.4838	-0.6822	-1.55
3.920	-0.4856	-0.6814	-1.55
3.925	-0.4874	-0.6804	-1.55
3.930	-0.4892	-0.6794	-1.54
3.935	-0.4907	-0.6782	-1.54
3.940	-0.4929	-0.6769	-1.54
3.945	-0.4946	-0.6759	-1.54
3.950	-0.4964	-0.6745	-1.54
3.955	-0.4981	-0.6732	-1.54
3.960	-0.5000	-0.6718	-1.54
3.965	-0.5018	-0.6704	-1.54
3.970	-0.5034	-0.6690	-1.54
3.975	-0.5055	-0.6677	-1.54
3.980	-0.5072	-0.6660	-1.54
3.985	-0.5090	-0.6647	-1.54
3.990	-0.5105	-0.6631	-1.55
3.995	-0.5124	-0.6616	-1.55
4.000	-0.5142	-0.6601	-1.55
4.005	-0.5157	-0.6586	-1.55
4.010	-0.5173	-0.6567	-1.56
4.015	-0.5189	-0.6552	-1.56
4.020	-0.5205	-0.6537	-1.56
4.025	-0.5220	-0.6520	-1.56

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	-0.5236	-0.6506	-1.56
4.035	-0.5254	-0.6488	-1.57
4.040	-0.5267	-0.6473	-1.57
4.045	-0.5281	-0.6455	-1.58
4.050	-0.5296	-0.6438	-1.58
4.055	-0.5308	-0.6421	-1.59
4.060	-0.5323	-0.6406	-1.59
4.065	-0.5333	-0.6392	-1.59
4.070	-0.5346	-0.6376	-1.60
4.075	-0.5359	-0.6362	-1.60
4.080	-0.5373	-0.6348	-1.60
4.085	-0.5381	-0.6334	-1.61
4.090	-0.5393	-0.6321	-1.61
4.095	-0.5407	-0.6310	-1.61
4.100	-0.5415	-0.6294	-1.62
4.105	-0.5427	-0.6285	-1.61
4.110	-0.5439	-0.6271	-1.62
4.115	-0.5452	-0.6261	-1.62
4.120	-0.5464	-0.6251	-1.62
4.125	-0.5477	-0.6240	-1.62
4.130	-0.5491	-0.6228	-1.61
4.135	-0.5504	-0.6214	-1.62
4.140	-0.5514	-0.6198	-1.62
4.145	-0.5521	-0.6184	-1.63
4.150	-0.5534	-0.6166	-1.63
4.155	-0.5540	-0.6151	-1.64
4.160	-0.5543	-0.6138	-1.65
4.165	-0.5543	-0.6124	-1.66
4.170	-0.5546	-0.6112	-1.67
4.175	-0.5540	-0.6104	-1.68
4.180	-0.5541	-0.6101	-1.68
4.185	-0.5542	-0.6092	-1.69
4.190	-0.5539	-0.6087	-1.69
4.195	-0.5537	-0.6084	-1.70
4.200	-0.5535	-0.6082	-1.70
4.205	-0.5536	-0.6080	-1.70
4.210	-0.5533	-0.6078	-1.70
4.215	-0.5528	-0.6081	-1.70
4.220	-0.5527	-0.6080	-1.71
4.225	-0.5524	-0.6081	-1.71
4.230	-0.5522	-0.6081	-1.71
4.235	-0.5520	-0.6083	-1.71
4.240	-0.5519	-0.6085	-1.71
4.245	-0.5515	-0.6092	-1.71
4.250	-0.5511	-0.6094	-1.71
4.255	-0.5509	-0.6099	-1.70

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	-0.5507	-0.6100	-1.70
4.265	-0.5500	-0.6104	-1.71
4.270	-0.5499	-0.6108	-1.70
4.275	-0.5494	-0.6115	-1.70
4.280	-0.5491	-0.6119	-1.70
4.285	-0.5488	-0.6123	-1.70
4.290	-0.5483	-0.6127	-1.70
4.295	-0.5479	-0.6132	-1.70
4.300	-0.5478	-0.6138	-1.70
4.305	-0.5478	-0.6143	-1.69
4.310	-0.5472	-0.6150	-1.69
4.315	-0.5471	-0.6155	-1.69
4.320	-0.5468	-0.6160	-1.69
4.325	-0.5465	-0.6168	-1.68
4.330	-0.5464	-0.6173	-1.68
4.335	-0.5461	-0.6181	-1.67
4.340	-0.5460	-0.6188	-1.67
4.345	-0.5459	-0.6197	-1.66
4.350	-0.5455	-0.6207	-1.66
4.355	-0.5451	-0.6215	-1.65
4.360	-0.5447	-0.6222	-1.65
4.365	-0.5444	-0.6235	-1.64
4.370	-0.5437	-0.6243	-1.64
4.375	-0.5434	-0.6253	-1.63
4.380	-0.5428	-0.6261	-1.63
4.385	-0.5422	-0.6268	-1.63
4.390	-0.5412	-0.6279	-1.63
4.395	-0.5409	-0.6285	-1.63
4.400	-0.5404	-0.6291	-1.63
4.405	-0.5398	-0.6297	-1.63
4.410	-0.5391	-0.6302	-1.62
4.415	-0.5384	-0.6305	-1.63
4.420	-0.5382	-0.6311	-1.62
4.425	-0.5374	-0.6315	-1.63
4.430	-0.5370	-0.6316	-1.63
4.435	-0.5370	-0.6314	-1.63
4.440	-0.5370	-0.6317	-1.63
4.445	-0.5366	-0.6319	-1.63
4.450	-0.5367	-0.6322	-1.63
4.455	-0.5365	-0.6329	-1.62
4.460	-0.5364	-0.6332	-1.62
4.465	-0.5360	-0.6346	-1.61
4.470	-0.5356	-0.6353	-1.61
4.475	-0.5352	-0.6364	-1.60
4.480	-0.5347	-0.6382	-1.59
4.485	-0.5348	-0.6397	-1.58

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	-0.5343	-0.6411	-1.57
4.495	-0.5342	-0.6423	-1.56
4.500	-0.5339	-0.6436	-1.55
4.505	-0.5335	-0.6446	-1.55
4.510	-0.5331	-0.6452	-1.55
4.515	-0.5324	-0.6454	-1.55
4.520	-0.5316	-0.6452	-1.56
4.525	-0.5311	-0.6459	-1.55
4.530	-0.5310	-0.6449	-1.56
4.535	-0.5308	-0.6445	-1.57
4.540	-0.5309	-0.6436	-1.57
4.545	-0.5311	-0.6424	-1.58
4.550	-0.5311	-0.6424	-1.58
4.555	-0.5316	-0.6419	-1.58
4.560	-0.5322	-0.6413	-1.58
4.565	-0.5321	-0.6409	-1.59
4.570	-0.5323	-0.6397	-1.59
4.575	-0.5322	-0.6389	-1.60
4.580	-0.5329	-0.6382	-1.60
4.585	-0.5334	-0.6374	-1.61
4.590	-0.5336	-0.6366	-1.61
4.595	-0.5341	-0.6358	-1.62
4.600	-0.5347	-0.6345	-1.62
4.605	-0.5352	-0.6338	-1.62
4.610	-0.5356	-0.6329	-1.63
4.615	-0.5362	-0.6322	-1.63
4.620	-0.5365	-0.6309	-1.64
4.625	-0.5367	-0.6299	-1.64
4.630	-0.5374	-0.6294	-1.64
4.635	-0.5374	-0.6284	-1.65
4.640	-0.5377	-0.6273	-1.66
4.645	-0.5379	-0.6266	-1.66
4.650	-0.5383	-0.6258	-1.67
4.655	-0.5382	-0.6247	-1.68
4.660	-0.5383	-0.6242	-1.68
4.665	-0.5385	-0.6230	-1.69
4.670	-0.5383	-0.6226	-1.69
4.675	-0.5381	-0.6215	-1.70
4.680	-0.5378	-0.6210	-1.71
4.685	-0.5374	-0.6201	-1.72
4.690	-0.5374	-0.6198	-1.72
4.695	-0.5369	-0.6192	-1.73
4.700	-0.5368	-0.6186	-1.73
4.705	-0.5368	-0.6175	-1.74
4.710	-0.5367	-0.6170	-1.75
4.715	-0.5367	-0.6162	-1.75

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.720	-0.5369	-0.6151	-1.76
4.725	-0.5373	-0.6142	-1.77
4.730	-0.5375	-0.6132	-1.77
4.735	-0.5379	-0.6123	-1.78
4.740	-0.5389	-0.6105	-1.78
4.745	-0.5398	-0.6095	-1.79
4.750	-0.5403	-0.6081	-1.79
4.755	-0.5414	-0.6067	-1.80
4.760	-0.5424	-0.6055	-1.80
4.765	-0.5428	-0.6048	-1.80
4.770	-0.5430	-0.6038	-1.81
4.775	-0.5434	-0.6029	-1.81
4.780	-0.5428	-0.6021	-1.82
4.785	-0.5427	-0.6017	-1.83
4.790	-0.5418	-0.6011	-1.84
4.795	-0.5413	-0.6003	-1.85
4.800	-0.5406	-0.5995	-1.86
4.805	-0.5396	-0.5989	-1.87
4.810	-0.5387	-0.5984	-1.88
4.815	-0.5374	-0.5977	-1.90
4.820	-0.5366	-0.5975	-1.90
4.825	-0.5364	-0.5972	-1.91
4.830	-0.5358	-0.5969	-1.91
4.835	-0.5353	-0.5970	-1.92
4.840	-0.5350	-0.5969	-1.92
4.845	-0.5350	-0.5971	-1.92
4.850	-0.5353	-0.5972	-1.92
4.855	-0.5351	-0.5977	-1.91
4.860	-0.5353	-0.5977	-1.91
4.865	-0.5357	-0.5987	-1.90
4.870	-0.5358	-0.5989	-1.90
4.875	-0.5365	-0.5995	-1.89
4.880	-0.5369	-0.6000	-1.88
4.885	-0.5374	-0.6004	-1.88
4.890	-0.5382	-0.6010	-1.87
4.895	-0.5382	-0.6014	-1.86
4.900	-0.5390	-0.6020	-1.85
4.905	-0.5394	-0.6023	-1.85
4.910	-0.5400	-0.6026	-1.84
4.915	-0.5407	-0.6031	-1.83
4.920	-0.5412	-0.6033	-1.83
4.925	-0.5418	-0.6035	-1.82
4.930	-0.5423	-0.6037	-1.81
4.935	-0.5428	-0.6041	-1.81
4.940	-0.5433	-0.6041	-1.80
4.945	-0.5439	-0.6044	-1.80

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	-0.5443	-0.6048	-1.79
4.955	-0.5447	-0.6049	-1.79
4.960	-0.5453	-0.6049	-1.78
4.965	-0.5458	-0.6052	-1.78
4.970	-0.5463	-0.6055	-1.77
4.975	-0.5468	-0.6058	-1.77
4.980	-0.5475	-0.6061	-1.76
4.985	-0.5479	-0.6061	-1.76
4.990	-0.5485	-0.6065	-1.75
4.995	-0.5490	-0.6067	-1.74
5.000	-0.5496	-0.6067	-1.74
5.005	-0.5498	-0.6069	-1.74
5.010	-0.5507	-0.6067	-1.73
5.015	-0.5510	-0.6069	-1.73
5.020	-0.5514	-0.6066	-1.73
5.025	-0.5515	-0.6066	-1.73
5.030	-0.5520	-0.6059	-1.73
5.035	-0.5515	-0.6063	-1.73
5.040	-0.5515	-0.6052	-1.74
5.045	-0.5513	-0.6058	-1.73
5.050	-0.5509	-0.6049	-1.74
5.055	-0.5505	-0.6058	-1.74
5.060	-0.5500	-0.6052	-1.75
5.065	-0.5502	-0.6059	-1.74
5.070	-0.5498	-0.6057	-1.74
5.075	-0.5507	-0.6059	-1.74
5.080	-0.5505	-0.6057	-1.74
5.085	-0.5514	-0.6057	-1.73
5.090	-0.5518	-0.6062	-1.73
5.095	-0.5533	-0.6052	-1.72
5.100	-0.5536	-0.6056	-1.72
5.105	-0.5549	-0.6053	-1.71
5.110	-0.5558	-0.6047	-1.71
5.115	-0.5566	-0.6038	-1.71
5.120	-0.5578	-0.6037	-1.70
5.125	-0.5588	-0.6026	-1.70
5.130	-0.5599	-0.6023	-1.70
5.135	-0.5610	-0.6013	-1.70
5.140	-0.5619	-0.6003	-1.70
5.145	-0.5630	-0.5990	-1.70
5.150	-0.5642	-0.5984	-1.70
5.155	-0.5652	-0.5969	-1.70
5.160	-0.5668	-0.5955	-1.70
5.165	-0.5676	-0.5940	-1.71
5.170	-0.5689	-0.5924	-1.71
5.175	-0.5703	-0.5906	-1.71

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	-0.5714	-0.5888	-1.72
5.185	-0.5726	-0.5874	-1.72
5.190	-0.5739	-0.5848	-1.73
5.195	-0.5749	-0.5830	-1.74
5.200	-0.5761	-0.5807	-1.74
5.205	-0.5770	-0.5789	-1.75
5.210	-0.5780	-0.5766	-1.76
5.215	-0.5792	-0.5744	-1.77
5.220	-0.5799	-0.5721	-1.78
5.225	-0.5812	-0.5700	-1.79
5.230	-0.5820	-0.5677	-1.80
5.235	-0.5830	-0.5654	-1.81
5.240	-0.5839	-0.5630	-1.82
5.245	-0.5850	-0.5608	-1.83
5.250	-0.5858	-0.5585	-1.84
5.255	-0.5869	-0.5562	-1.84
5.260	-0.5879	-0.5541	-1.85
5.265	-0.5888	-0.5523	-1.86
5.270	-0.5898	-0.5502	-1.87
5.275	-0.5908	-0.5480	-1.88
5.280	-0.5917	-0.5461	-1.88
5.285	-0.5924	-0.5441	-1.89
5.290	-0.5935	-0.5422	-1.90
5.295	-0.5941	-0.5405	-1.90
5.300	-0.5951	-0.5387	-1.91
5.305	-0.5960	-0.5369	-1.91
5.310	-0.5968	-0.5351	-1.92
5.315	-0.5975	-0.5336	-1.93
5.320	-0.5979	-0.5317	-1.94
5.325	-0.5987	-0.5304	-1.94
5.330	-0.5993	-0.5286	-1.95
5.335	-0.6000	-0.5267	-1.96
5.340	-0.6005	-0.5254	-1.96
5.345	-0.6012	-0.5239	-1.97
5.350	-0.6014	-0.5223	-1.98
5.355	-0.6021	-0.5203	-1.98
5.360	-0.6025	-0.5183	-2.00
5.365	-0.6030	-0.5169	-2.00
5.370	-0.6038	-0.5150	-2.01
5.375	-0.6045	-0.5132	-2.01
5.380	-0.6052	-0.5117	-2.02
5.385	-0.6065	-0.5102	-2.02
5.390	-0.6071	-0.5088	-2.02
5.395	-0.6078	-0.5073	-2.03
5.400	-0.6086	-0.5061	-2.03
5.405	-0.6097	-0.5050	-2.03

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.410	-0.6107	-0.5036	-2.03
5.415	-0.6114	-0.5021	-2.04
5.420	-0.6122	-0.5011	-2.03
5.425	-0.6135	-0.5001	-2.03
5.430	-0.6137	-0.4988	-2.04
5.435	-0.6147	-0.4980	-2.04
5.440	-0.6157	-0.4971	-2.03
5.445	-0.6162	-0.4962	-2.03
5.450	-0.6170	-0.4954	-2.03
5.455	-0.6175	-0.4950	-2.03
5.460	-0.6183	-0.4941	-2.03
5.465	-0.6187	-0.4934	-2.03
5.470	-0.6196	-0.4923	-2.03
5.475	-0.6198	-0.4916	-2.04
5.480	-0.6209	-0.4908	-2.03
5.485	-0.6215	-0.4900	-2.03
5.490	-0.6224	-0.4892	-2.03
5.495	-0.6230	-0.4886	-2.03
5.500	-0.6239	-0.4881	-2.02
5.505	-0.6249	-0.4879	-2.02
5.510	-0.6257	-0.4873	-2.01
5.515	-0.6267	-0.4870	-2.01
5.520	-0.6276	-0.4869	-2.00
5.525	-0.6283	-0.4870	-1.99
5.530	-0.6294	-0.4871	-1.98
5.535	-0.6300	-0.4870	-1.98
5.540	-0.6309	-0.4872	-1.97
5.545	-0.6316	-0.4875	-1.96
5.550	-0.6321	-0.4876	-1.96
5.555	-0.6327	-0.4876	-1.95
5.560	-0.6335	-0.4876	-1.94
5.565	-0.6343	-0.4877	-1.94
5.570	-0.6351	-0.4879	-1.93
5.575	-0.6358	-0.4879	-1.92
5.580	-0.6367	-0.4880	-1.91
5.585	-0.6375	-0.4879	-1.91
5.590	-0.6383	-0.4880	-1.90
5.595	-0.6393	-0.4880	-1.89
5.600	-0.6402	-0.4880	-1.88
5.605	-0.6412	-0.4882	-1.87
5.610	-0.6423	-0.4884	-1.86
5.615	-0.6433	-0.4886	-1.85
5.620	-0.6441	-0.4889	-1.84
5.625	-0.6452	-0.4894	-1.83
5.630	-0.6459	-0.4896	-1.83
5.635	-0.6467	-0.4898	-1.82

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.6472	-0.4898	-1.81
5.645	-0.6477	-0.4902	-1.81
5.650	-0.6486	-0.4903	-1.80
5.655	-0.6492	-0.4904	-1.79
5.660	-0.6498	-0.4905	-1.79
5.665	-0.6502	-0.4906	-1.78
5.670	-0.6511	-0.4906	-1.77
5.675	-0.6513	-0.4906	-1.77
5.680	-0.6523	-0.4906	-1.76
5.685	-0.6521	-0.4907	-1.77
5.690	-0.6523	-0.4903	-1.77
5.695	-0.6525	-0.4899	-1.77
5.700	-0.6530	-0.4898	-1.76
5.705	-0.6531	-0.4895	-1.76
5.710	-0.6538	-0.4892	-1.76
5.715	-0.6534	-0.4889	-1.77
5.720	-0.6539	-0.4888	-1.76
5.725	-0.6543	-0.4886	-1.76
5.730	-0.6547	-0.4880	-1.76
5.735	-0.6550	-0.4880	-1.76
5.740	-0.6553	-0.4875	-1.76
5.745	-0.6549	-0.4879	-1.76
5.750	-0.6551	-0.4867	-1.76
5.755	-0.6553	-0.4861	-1.77
5.760	-0.6551	-0.4856	-1.77
5.765	-0.6552	-0.4852	-1.77
5.770	-0.6554	-0.4841	-1.78
5.775	-0.6554	-0.4832	-1.79
5.780	-0.6552	-0.4823	-1.79
5.785	-0.6553	-0.4812	-1.80
5.790	-0.6555	-0.4804	-1.80
5.795	-0.6551	-0.4793	-1.81
5.800	-0.6548	-0.4780	-1.82
5.805	-0.6551	-0.4770	-1.83
5.810	-0.6549	-0.4758	-1.84
5.815	-0.6548	-0.4742	-1.85
5.820	-0.6547	-0.4732	-1.85
5.825	-0.6548	-0.4717	-1.86
5.830	-0.6545	-0.4704	-1.87
5.835	-0.6542	-0.4692	-1.88
5.840	-0.6542	-0.4677	-1.89
5.845	-0.6543	-0.4665	-1.90
5.850	-0.6541	-0.4654	-1.91
5.855	-0.6541	-0.4637	-1.92
5.860	-0.6543	-0.4624	-1.93
5.865	-0.6544	-0.4610	-1.93

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.6542	-0.4596	-1.94
5.875	-0.6545	-0.4583	-1.95
5.880	-0.6547	-0.4572	-1.95
5.885	-0.6548	-0.4558	-1.96
5.890	-0.6551	-0.4548	-1.97
5.895	-0.6549	-0.4537	-1.97
5.900	-0.6549	-0.4528	-1.98
5.905	-0.6552	-0.4516	-1.98
5.910	-0.6553	-0.4510	-1.99
5.915	-0.6551	-0.4499	-2.00
5.920	-0.6549	-0.4488	-2.00
5.925	-0.6548	-0.4478	-2.01
5.930	-0.6546	-0.4471	-2.02
5.935	-0.6546	-0.4458	-2.03
5.940	-0.6546	-0.4447	-2.03
5.945	-0.6546	-0.4441	-2.04
5.950	-0.6547	-0.4428	-2.04
5.955	-0.6549	-0.4419	-2.05
5.960	-0.6553	-0.4409	-2.05
5.965	-0.6557	-0.4398	-2.05
5.970	-0.6562	-0.4395	-2.05
5.975	-0.6564	-0.4387	-2.05
5.980	-0.6570	-0.4388	-2.05
5.985	-0.6572	-0.4386	-2.05
5.990	-0.6580	-0.4383	-2.04
5.995	-0.6583	-0.4384	-2.04
6.000	-0.6587	-0.4388	-2.03

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable (|S11_ref1-S11_ref2| < 1e-02). Consider starting the sweep above the affected range.

- 5 frequency point(s) with $\varepsilon'' < 0$ (non-passive, worst ε'' = -0.702), attributable to measurement noise on a near-lossless liquid. Values kept and flagged, not replaced by another root.